

- eliminate a migrated process's residual dependencies on its original node.
- earliest-deadline-first (EDF)**—Scheduling policy that gives a processor to the process with the closest deadline.
- ease of use**—Measure of the comfort and convenience associated with system use.
- EBCDIC (Extended Binary-Coded Decimal Interchange Code)**—Eight-bit character set for representing data in mainframe computer systems, particularly systems developed by IBM.
- effective priority (Linux)**—Priority assigned to a process by adding its static priority to its priority boost or penalty.
- efficient operating system**—Operating system that exhibits high throughput and small turnaround time.
- elevator algorithm**—See SCAN disk scheduling.
- embedded system**—Small computer containing limited resources and specialized hardware to run devices such as PDAs or cellular phones.
- emulation stage**—When a Coda client becomes disconnected, it is said to enter the emulation stage. During this stage all file read requests are satisfied from cache. Write requests during the emulation stage occur in two steps. First, the file is updated on disk. Second, a log called the client modification log (CML) is updated to reflect file changes.
- encapsulating security payload (ESP) (IPSec)**—Message data encrypted using symmetric-key ciphers to protect the data from eavesdroppers while the IP packet is transmitted across public communication lines.
- Encrypting File System (EFS)**—NTFS feature that uses cryptography to protect files and folders in Windows XP Professional and Windows 2000. EFS uses secret-key and public-key encryption to secure files. Each user is assigned a key pair and certificate that are used to ensure that only the user that encrypted the files can access them.
- encryption**—Technique that transforms data to prevent it from being interpreted by unauthorized users.
- entry queue**—See entry set.
- entry set**—In Java, a queue of threads waiting to enter a monitor after calling a synchronized method.
- environment subsystem (Windows XP)**—User-mode component that provides a computing environment for other user-mode processes; in most cases, only environment subsystems interact directly with kernel-mode components in Windows XP.
- epoch (Linux)**—Time during which all processes move from the scheduler's active list to its expired list. This ensures that processes are not indefinitely postponed.
- Ethernet (IEEE 802.3)**—Network that supports many speeds over a variety of cables. Ethernet uses the Carrier Sense Multiple Access with Collision Detection (CSMA/CD) protocol. Ethernet is the most popular type of LAN.
- event-driven simulator**—Simulator controlled by events that are made to occur according to probability distributions.
- event object (Windows XP)**—Synchronization object that becomes *signaled* when a particular event occurs.
- exception**—Error caused by a process. Processor exceptions invoke the operating system, which determines how to respond. Processes can register exception handlers that are executed when the operating system receives the corresponding exception.
- exception (Intel IA-32 specification)**—Hardware signal generated by an error. In the Intel IA-32 specification, exceptions are classified as traps, faults and aborts.
- execute access (virtual memory)**—Access right that enables a process to execute instructions from a page or segment; see also read access, write access and append access.
- execute access (file)**—Permission that enables a user to execute a file.
- execution mode**—Operating system execution mode (e.g., user mode or kernel mode) that determines which instructions can be executed by a process.
- executive (Windows XP)**—Portion of the Windows XP operating system that is responsible for managing the operating system's subsystems (e.g., I/O subsystem, memory subsystem and file system).
- executive mode**—Protected mode in which a processor can execute operating system instructions on behalf of a user (also called kernel mode).
- executive process (EPROCESS) block (Windows XP)**—Executive data structure that stores information about a process, such as that process's object handles and ID; an EPROCESS block also stores the process's KPROCESS block.
- executive resource lock (Windows XP)**—Synchronization lock available only to kernel-mode threads and that may be held either in shared mode by many threads or in exclusive mode by one thread.
- executive thread (ETHREAD) block (Windows XP)**—Executive data structure that stores information about a thread, such as the thread's pending I/O requests and the thread's start address; an ETHREAD block also stores the thread's KTHREAD block.
- expected value**—Sum of a series of values each multiplied by its respective probability of occurrence.
- expired list (Linux)**—Structure containing processes that cannot contend for the processor until the next epoch. Processes are placed in this list to prevent others from being indefinitely postponed. To quickly begin a new epoch, this list becomes the active list.
- expired state (Linux)**—Task state that prevents a task from being dispatched until the next epoch.
- explicit acknowledgement**—The client sends an acknowledgement to the server in an additional packet when the client receives the response from the server.
- explicit handle (Windows XP)**—Binding handle in which the client must specify all binding information and create and manage the handle.
- Explicitly Parallel Instruction Computing (EPIC)**—Processor-design philosophy whose goals are to provide a high degree of instruction-level parallelism, reduce processor hardware complexity and improve performance.

- exponential backoff**—Method employed by Ethernet to calculate the interval before retransmission after a collision; this reduces the chance of subsequent collisions on the same transmission, thus increasing throughput.
- export local file**—Performed by an NFS server to make the local directory of files available to the remote client via the mount protocol.
- ext2 inode** (Linux)—Structure that stores information such as file size, the location of a file's data blocks and permissions for a single file or directory in an ext2 file system.
- ext2fs** (Linux)—Popular inode-based Linux file system that enables fast access to small files and supports large file sizes.
- Extensible Firmware Interface (EFI)**—Interface designed by Intel that improves upon a traditional BIOS by supporting device drivers and providing a shell interface at boot time.
- Extensible Markup Language (XML)**—Language for marking up data so that information can be exchanged between applications and platforms.
- extensible operating system**—An operating system that can incorporate new features easily.
- extent**—Block of contiguous sectors.
- external fragmentation**—Phenomenon in variable-partition memory systems in which there are holes distributed throughout memory that are too small to hold a process.
- external name**—Symbol defined in a module that can be referenced by other modules.
- external reference**—Reference from one module to an external name in a different module.

F

- fabric layer** (in grid computing)—Layer that accesses physical resources, such as disks.
- failure transparency**—Method by which a distributed system provides fault tolerance so that the client is unaware of the failure of a computer.
- fair share group**—Group of processes that receives a percentage of the processor time under a fair share scheduling (FSS) policy.
- fair share scheduling (FSS)**—Scheduling policy developed for AT&T's UNIX system that places processes in groups and assigns these groups a percentage of processor time.
- fairness**—Property of a scheduling algorithm that treats all processes equally.
- false sharing**—Situation that occurs when processes on separate processors are forced to share a page because they are each accessing a data item on that page, although not the same data item.
- family of computers**—Series of computers that are compatible in that they can run the same programs.
- far page-replacement strategy**—Graph-based page-replacement strategy that analyzes a program's reference patterns to determine which page to replace. This strategy replaces the page that is furthest from any referenced page in the graph and that has not been referenced recently.
- fast instruction set computing (FISC)**—Term describing the processor-design philosophy resulting from the convergence of RISC and CISC design philosophies. The FISC design philosophy stresses inclusion of any construct that improves performance.
- fast mutex** (Windows XP)—Efficient mutex variant that operates at the APC level with some restrictions (e.g., a thread cannot specify a maximum wait time to wait for a fast mutex).
- fast mutual exclusion algorithm**—Implementation of mutual exclusion that avoids the overhead of a thread performing multiple tests when no other thread is contending for its critical section. This first fast mutual exclusion algorithm was proposed by Lamport.
- fast user switching** (Windows XP)—Ability of a new user to log onto a Windows XP machine without logging off the previous user.
- FAT file system**—An implementation of tabular noncontiguous file allocation developed by Microsoft.
- fault**—In the Intel IA-32 specification, an exception as the result of an error such as division by zero or illegal access to memory. Some faults can be corrected by appropriate operating system exception handlers.
- fault tolerance**—Operating system's ability to handle software or hardware errors.
- feature freeze** (Linux)—State of kernel development during which no new features should be added to the kernel, in preparation for a new kernel release.
- feedback loop**—Technique in which information about the current state of the system can influence the number of requests arriving at a resource (e.g., positive and negative feedback loops).
- fetch strategy**—Method of determining when to obtain the next piece of program or data for transfer from secondary storage to main memory.
- fiber** (Windows XP)—Unit of execution in Windows XP created by a thread and scheduled by that thread. Fibers facilitate portability for applications that schedule their own threads.
- Fiber Distributed Data Interface (FDDI)**—Protocol that shares many properties of a Token Ring, but operates over fiber-optic cable, allowing the transfer of more information at greater speeds. In FDDI, a token circulates around the optical fiber ring; stations cannot transmit until they obtain the token by receiving it from a preceding station. FDDI uses a second Token Ring as backup or to circulate tokens in the reverse direction of the primary Token Ring.
- fiber local storage (FLS)** (Windows XP)—Area of a process's address space where a fiber can store data that only the fiber can access.
- field**—In the data hierarchy, a group of characters (e.g., a person's name, street address or telephone number).
- FIFO** (Linux)—Named pipe that enables two unrelated processes to communicate via the producer/consumer relationship using a page-size buffer.
- FIFO anomaly**—Phenomenon in FIFO page-replacement strategy whereby increasing a process's page frame allocation increases

the number of page faults it experiences; normally, page faults should decrease as more page frames become available.

FIFO broadcast—Guarantees that when two messages are sent from one process to another, the message that was sent first will arrive first.

file—Named collection of data that may be manipulated as a unit by operations such as open, close, create, destroy, copy, rename and list. Individual data items within a file may be manipulated by operations like read, write, update, insert and delete. File characteristics include location, accessibility, type, volatility and activity. Files can consist of one or more records.

file allocation table (FAT)—Table storing pointers to file data blocks in Microsoft's FAT file system.

file attribute (Linux)—File metadata that implements access control information, such as whether a file is append-only or immutable, that cannot be specified using standard Linux file permissions.

file control block—Metadata containing information the system needs to manage a file, such as access control information.

file descriptor—Non-negative integer that indexes into an opened-file table. A process references a file descriptor instead of a pathname to access file data without incurring the overhead of a directory structure traversal.

file handle—Identifies a file on the file server with file type, location and access permissions.

file identifiers (fids) (in AFS)—Entity that specifies a volume, an index within a volume and an identifier to guarantee object uniqueness within a volume.

file integrity mechanism—Mechanism that ensures that the information in a file is uncorrupted. When file integrity is assured, files contain only the information they are intended to have.

file management—Component of a file system concerned with providing the mechanisms for files to be stored, referenced, shared and secured.

file mapping (Windows XP)—Interprocess communication mechanism in which multiple processes access the same file by placing it in their virtual memory space. The different virtual addresses correspond to the same main memory addresses.

file mapping object (Windows XP)—Object used by processes to map any file into memory.

file organization—Manner in which the records of a file are arranged on secondary storage (e.g., sequential, direct, indexed sequential and partitioned).

file permission—Structure that determines whether a user may read, write and or execute a file.

file server—System dedicated to provide remote processes access to its files.

file system—Component of an operating system that organizes files and manages access to data. File systems are concerned with organizing files logically (using pathnames) and physically (using metadata). They also manage their storage device's free space, enforce security policies, maintain data integrity and so on.

file system identifier—Value that uniquely identifies the file system a storage device is using.

file system manager—Operating system component that organizes named collections of data on storage devices and provides an interface for accessing data on those devices.

File Transfer Protocol (FTP)—Application layer protocol that moves files between different hosts on a network. FTP specifies connections between two pairs of ports: one pair sends control information that governs the session, the other sends the actual data.

file view (Windows XP)—Portion of a file specified by a file mapping object that a process maps into its memory.

filter driver—WDM (Windows Driver Model) driver that modifies the behavior of a hardware device (e.g., providing mouse acceleration) or adds some extra services (e.g., security checks).

fine-grained strip (RAID)—Strip size that causes average files to be stored in multiple stripes. Fine-grained strips can reduce each request's access time and increase transfer rates because multiple disks simultaneously retrieve portions of the requested data.

firewall—Software or hardware that protects a local area network from packets sent by malicious users from an external network.

firmware—Microcode that specifies simple, fundamental instructions necessary to implement machine-language instructions.

first-come-first-served (FCFS) disk scheduling—Disk scheduling strategy in which the earliest arriving request is serviced first. FCFS is a fair method of allocating service, but when the request rate (i.e., the load) becomes heavy, FCFS can result in long waiting times. FCFS exhibits a random seek pattern in which successive requests can cause time-consuming seeks from the innermost to the outermost cylinders.

first-come-first-served (FCFS) process scheduling—Job-blind multiprocessor scheduling algorithm that places arriving processes in a queue; the process at the head of the queue executes until it freely relinquishes the processor.

first-fit memory placement strategy—Memory placement strategy that places an incoming process in the first hole that is large enough to hold it.

first-in-first-out (FIFO)—Nonpreemptive scheduling policy that dispatches processes according to their arrival time in the ready queue.

first-in-first-out (FIFO) page replacement—Page-replacement strategy that replaces the page that has been in memory longest. FIFO incurs low overhead but generally does not predict future page usage accurately.

fixed-partition multiprogramming—Memory organization that divides main memory into a number of fixed-size partitions, each holding a single job.

flat directory structure—File system organization containing only one directory.

flow control—Means by which TCP regulates the number of segments sent by a host to avoid overwhelming the receiver.

flushing—Process migration strategy in which the sending node writes all of the process's memory pages to a shared disk at the start of

migration; the process then accesses the pages from the shared disk as needed on the receiving node.

flushing a page—Copying the contents of a modified page in main memory to secondary storage so another page can be placed in its frame. When this occurs, the page's dirty bit is cleared, which enables the operating system to quickly determine that the page can be overwritten by an incoming page, which can reduce page-wait times.

format a storage device—To prepare a device for a file system by performing operations such as inspecting its contents and writing storage management metadata.

Fortran—Procedural programming language developed by IBM in the mid-1950s for scientific applications that require complex mathematical computations.

fragmentation (of main memory)—Phenomenon wherein a system is unable to make use of certain areas of available main memory.

fragmented disk—Disk that stores files in discontinuous blocks as the result of file creation and deletion. Such disks exhibit high seek times when reading files sequentially. Disk defragmentation can reduce or eliminate the problem.

frame—Piece of data in the link layer. Contains both the message and the control information.

free list—Linked list of blocks that contain the addresses of free blocks.

free memory list—Operating system data structure that points to available holes in memory.

free model—COM (Microsoft's Component Object Model) object threading model in which many threads can act as the server for a COM object.

Free Page List (Windows XP)—List of page frames that are available for reclaiming; although these page frames do not contain any valid data, they may not be used until the zero-page thread sets all of their bits to zero.

frontside bus (FSB)—Bus that connects a processor to main memory.

FSCAN disk scheduling—Disk scheduling strategy that uses SCAN to service only those requests waiting when a particular sweep begins (the "F" stands for "freezing" the request queue at a certain time). Requests arriving during a sweep are grouped together and ordered for optimum service during the return sweep.

fully-connected mesh network—Mesh network in which each node is directly connected to every other node. These networks are faster and more fault tolerant than other networks, but also unrealizable on all but the smallest of networks because of the cost of the potentially enormous number of connections.

function driver—WDM (Windows Driver Model) device driver that implements a device's main functions; it does most of the I/O processing and provides the device's interface.

G

gang scheduling—Another name for rescheduling.

garbage collection—See memory compaction.

general protection fault (GPF) (IA-32 Intel architecture)—Occurs when a process references a segment to which it does not have appropriate access rights or references an address outside of the segment.

General Public License (GPL)—Open-source software license which specifies that software distributed under it must contain the complete source code, must clearly indicate any modifications to the original code and must be accompanied by the GPL. End users are free to modify and redistribute any software under the GPL.

general-purpose register—Register that can be used by processes to store data and pointer values. Special-purpose registers cannot be accessed by user processes.

general semaphore—See counting semaphore.

global descriptor table (GDT) (IA-32 Intel architecture)—Segment map table that contains mapping information for process segments or local descriptor table (LDT) segments, which contain mapping information for process segments.

global least-recently-used (gLRU) page replacement—Global page-replacement strategy that replaces the page that has not been referenced for the longest time in the entire system. LRU can perform poorly because variants of round-robin scheduling cause the system to exhibit a large-scale looping reference pattern. The SEQ variant of gLRU page replacement attempts to improve performance by replacing the most-recently-used page when it detects a looping reference pattern.

global run queue—Process scheduling queue used in some multiprocessor scheduling algorithms, which is independent of the processors in a system and into which every process or job in the system is placed.

globally unique ID (GUID)—128-bit integer that is, for all practical purposes, guaranteed to be unique in the world. COM (Microsoft's Component Object Model) uses GUIDs to uniquely identify interfaces and object classes.

GNU—Project initiated by Stallman in the 1980s aimed at producing an open-source operating system with the features and utilities of UNIX.

granularity bit (IA-32 Intel architecture)—Bit that determines how the processor interprets the size of each segment, specified by the 20-bit segment limit. When the bit is off, segments range in size from 1 byte to 1MB, in 1-byte increments. When the bit is on, segments range in size from 4KB to 4GB, in 4KB increments.

graph reduction—Altering a resource-allocation graph by removing a process if that process can complete. This also involves removing any arrows leading to the process (from the resources allocated to the process) or away from the process (to resources the process is requesting). A resource-allocation graph can be reduced by a process if all of that process's resource requests can be granted, enabling that process to run to completion and free its resources.

Graphical Identification and Authentication (GINA) (Windows XP)—Graphical user interface that prompts users for credentials, usually in the form of a username and password. Windows XP ships with its own implementation of GINA called MSCINA, but it accepts third-party DLLs.

graphical user interface (GUI)—User-friendly point of access to an operating system that uses graphical symbols such as windows, icons and menus to facilitate program and file manipulation.

grid computing—Links computational resources that are distributed over the wide area network (such as computers, data storages and scientific devices) to solve complex problems.

group (file access control)—Set of users with the same file access rights (e.g., members of a group that is working on a particular project).

group descriptor (Linux) — Structure that records information regarding a block group, such as the locations of the inode allocation bitmap, block allocation bitmap and inode table.

H

hacker—Experienced programmer who often program as much for personal enjoyment as for the functionality of the application. This term is often used when the term cracker is more appropriate.

Hamming error-correcting codes (Hamming ECCs)—Technique of generating parity bits that enables systems to detect and correct errors in data transmission.

handshaking—Mechanism in a connection-oriented transport layer in which hosts send control information to create a logical connection between the hosts.

happens before relation—A happens before B if A and B belong to the same process and A occurred before B; or A is the sending of a message and B is the receiving of that message.

hard affinity—Type of processor affinity in which the scheduling algorithm guarantees that a process only executes on a single node throughout its life cycle.

hard disk drive—Magnetic, rotational secondary storage device that provides persistent storage for, and random access to, data.

hard link—Directory entry specifying the location of a file on its storage device.

hard real-time scheduling—Scheduling policy that ensures processes meet their deadlines.

Hardware Abstraction Layer (HAL) (Windows XP)—Operating system component that interacts directly with the hardware and abstracts hardware details for other system components.

Hartstone—Popular synthetic benchmark used to evaluate real-time systems.

hash anchor table—Hash table that points to entries in an inverted page table. Increasing the size of the hash anchor table decreases the number of collisions, which improves the speed of address translation, at the cost of the increased memory overhead required to store the table.

hash function—Function that takes a number as input and returns a number, called a hash value, within a specified range. Hash functions facilitate rapidly storing and retrieving information from hash tables.

hash table—Data structure that indexes items according to their hash values; used with hash functions to rapidly store and retrieve information.

hash value—Value returned by a hash function that corresponds to a position in a hash table.

Havender's linear ordering—See linear ordering.

hbench microbenchmark suite—Popular microbenchmark suite, which enables evaluators to effectively analyze the relationship between operating system primitives and hardware components.

head node (in a Beowulf cluster)—Node, also called master node, that acts as a server to distribute the workload, control access to the cluster and handle the shared resources.

header—Control information placed in front of a data message.

heavyweight process (HWP)—A traditional process, which may contain one or more threads. The process is "heavyweight" because it is allocated its own address space upon creation.

heuristic scanning—Antivirus technique that detects viruses by their program behavior.

heuristics—Technique that solves complex problems using rules of thumb or other approximations that incur low execution overhead and generally provide good results.

hierarchical deadlock detection—A strategy in distributed deadlock, which arranges each site in the system as a node in a tree. Each node, except the leaf nodes, collects the resource allocation information of all dependent nodes.

hierarchical process structure—Organization of processes when parent processes spawn child processes and, in particular, only one parent creates a child.

hierarchically structured file system—File system organization in which each directory can contain multiple subdirectories but exactly one parent.

high-availability cluster—Cluster in which only some of the nodes are working while other nodes act as backups. The goal of a high-availability cluster is to stay up all the time.

high IRQ (Windows XP)—Highest-priority IRQ, at which machine check and bus error interrupts execute and all other interrupts are masked.

high-level driver (Windows XP)—Device driver that abstracts hardware specifics and passes I/O requests to low-level drivers.

high-level language—Programming language that uses English-like identifiers and common mathematical notation to represent programs using fewer statements than assembly-language programming.

high-level scheduling—Determining which jobs a system allows to compete actively for system resources.

high memory (Linux)—Region of physical memory (which begins at 896MB on the IA-32 architecture) beginning at the largest physical address that is permanently mapped to the kernel's virtual address space and extending to the limit of physical memory (64GB on Intel Pentium 4 processors). Because the kernel must perform expensive operations to map pages in its virtual address space to page frames in high memory, most kernel data structures are not stored in high memory.

high-performance cluster—Cluster in which all the nodes work to achieve maximum performance.

highest-response-ratio-next (HRRN)—Scheduling policy that assigns priority based on a process's service time and the amount of time the process has been waiting.

hoarding stage (in Coda)—Stage that clients enter when they are connected to Coda. In this stage, clients prepare for a possible disconnection from the system by caching any requested file.

hole—An unused area of memory in a variable-partition multiprogramming system.

home-based consistency—Memory-coherence strategy in which processors send coherence information to a home node associated with the page being written; the home node forwards update information to other nodes that subsequently access the page.

home computer (in process migration)—Computer on which the process originates.

home node—Node that is the "home" for a physical memory address or page and is responsible for maintaining the data's coherence.

hook—Software feature that enables developers to add features to an existing application without modifying its source file. An application uses a hook to call a procedure that can be defined by another application.

host—Entity, such as a computer or Internet-enabled cellular phone, that receives and/or provides services over a network. Also called a node.

host-based intrusion detection—IDS that monitors system and application log files.

hot spare disk (RAID)—Disk in a RAID system that is not used until a disk fails. Once the system regenerates the failed disk's data, the hot spare disk replaces the failed disk.

hot spot—Disk cylinder that contains frequently requested data. Some disk arm anticipation techniques move the disk head to hot spots to reduce average seek times.

hot swappable device—Device that can be added to, or removed from, a computer while it is running.

HTTP request—Resource request from an HTTP client to an HTTP server.

HTTP response—Reply message from an HTTP server to an HTTP client, consisting of a status, header and data.

hub—Central node (such as in a star network) responsible for relaying messages between nodes.

hybrid methodology—Performance evaluation technique that combines the vector-based methodology with trace data to measure performance for applications whose behavior depends strongly on user input.

hypercube—Multiprocessor interconnection scheme that consists of $2n$ nodes (where n is an integer); each node is linked with n neighbor nodes.

HyperText Markup Language (HTML)—Language that specifies the content and arrangement of information on a Web page and provides hyperlinks to access other pages.

Hypertext Transfer Protocol (HTTP)—Application layer protocol used for transferring HTML documents and other data formats

between a client and a server. This is the key protocol of the World Wide Web.

I

I/O-bound—Process (or job) that tends to use a processor for a short time before generating an I/O request and relinquishing the processor.

I/O channel—Component responsible for handling device I/O independently of a main processor.

I/O completion interrupt—Message issued by a device when it finishes servicing an I/O request.

I/O completion port (Windows XP)—Port at which threads register and block, waiting to be awakened when processing completes on an I/O request.

I/O completion routine (Windows XP)—Function registered by a device driver with the I/O manager in reference to an IRP; the I/O manager calls this function when processing of the IRP completes.

I/O manager—Operating system component that receives, interprets and performs I/O requests.

I/O manager (Windows XP)—Executive component that interacts with device drivers to handle I/O requests.

I/O request packet (IRP) (Windows XP)—Data structure that describes an I/O request.

I/O status block (Windows XP)—Field in an IRP that indicates whether an I/O request completed successfully or, if not, the request's error code.

I/O throttling (Windows XP)—Technique that increases stability when available memory is low; the VMM manages memory one page at a time during I/O throttling.

IBSYS—Operating system for the IBM 7090 mainframe.

ideal processor (Windows XP)—Thread attribute that specifies a processor on which the system should attempt to schedule the thread for execution.

idempotent request—A requested operation that if performed several times will return the same result, so it is acceptable to perform the same operation twice.

IEEE 1394 port—Commonly used serial port that provides transfer speeds of up to 800MB per second, sometimes supplies power to devices and allows devices to be hot swappable; these ports are commonly referred to as Fire Wire (from Apple) or iLink (from Sony).

IEEE 802.11—One of the standards that governs wireless communication. It dictates that hosts follow the CSMA/CA protocol.

immutable attribute (Linux)—Attribute specifying that a file can be read and executed, but cannot be copied, modified or deleted.

immutable file—A file that cannot be modified after it is created.

implicit acknowledgement—The response message implies the acknowledgement for the request message.

inactive list (Linux)—See expired list.

- inactive page** (Linux)—Page in main memory that can be replaced by an incoming page.
- incremental backup**—Backup technique that copies only data in the file system that has changed since the last backup.
- indefinite postponement**—Situation in which a thread waits for an event that might never occur.
- independent software vendor (ISV)**—Organization that develops and sells software. ISVs prospered after the release of the IBM PC.
- index block**—Block that contains a list of pointers to file data blocks.
- indexed sequential file organization**—File organization that arranges records in a logical sequence according to a key contained in each record.
- indirect block**—Index block containing pointers to data blocks in inode-based file systems.
- indirect pointer**—Inode pointer that points to a block of inode pointers.
- indivisible operation**—See atomic operation.
- information hiding**—Software architectural technique that facilitates the development of more reliable software systems by preventing direct access to data within an object by outside objects.
- inheritance** (Windows XP)—Technique by which a child process obtains attributes (e.g., most types of handles and the current directory) from its parent process upon creation.
- initialized state** (Windows XP)—Thread state in which the thread is created by the operating system.
- inode**—Index block in a UNIX-based system that contains the file control block and pointers to singly, doubly and triply indirect blocks of pointers to file data.
- inode** (Linux) — Structure that describes the location of data blocks corresponding to a file, directory or link in a file system. In the VFS, this structure represents any file in the system. An ext2 inode represents a file in the ext2 file system.
- inode allocation bitmap** (Linux)—Bitmap that records a block group's inode usage.
- inode cache** (Linux)—Cache that improves inode lookup performance.
- inode map**—Block of metadata written to the log of a log-structured file system that indicates the location of the file system's inodes. Inode maps improve LFS performance by reducing the time required to determine file locations in the LFS.
- inode table** (Linux)—Structure that contains an entry for each allocated inode in a block group.
- input/output control system (IOCS)**—Precursor to modern operating systems that provided programmers with a basic set of functions to perform I/O.
- insert** (file)—Operation that adds a new data item to a file.
- instruction decode unit**—Component of a processor that interprets instructions and generates appropriate control signals that cause the processor to perform each instruction.
- instruction fetch unit**—Component of a processor that loads instructions from the instruction cache so they can be decoded and executed.
- instruction length**—Number of bits that comprise an instruction in a given architecture. Some architectures support variable-length instructions; instruction lengths also vary among different architectures.
- instruction-level parallelism (ILP)**—Parallelism that permits two machine instructions to be executed at once. Two instructions exhibit ILP if the execution of one does not affect the outcome of the other (i.e., the two instructions do not depend on each other).
- instruction set**—Set of machine instructions a processor can perform.
- instruction set architecture (ISA)**—Interface exposed by a processor that describes the processor, including its instruction set, number of registers and memory size.
- instrumentation level** (in JMX)—Makes any Java-based object manageable so that the management application can access and operate these objects.
- integrity** (secure transaction)—One of the five fundamental requirements for a successful, secure transaction. Integrity deals with how to ensure that the information you send or receive has not been compromised or altered.
- intensive resource management**—Notion of devoting substantial resources to managing other resources to improve overall utilization.
- interactive operating system**—Operating system that allows applications to respond quickly to user input.
- Interactive Problem Control System (IPCS)** (VM)—VM component that provides online analysis and correction of VM software problems.
- interactive process**—Process that requires user input as it executes.
- interactive users**—Users that are present when the system processes their jobs. Interactive users communicate with their jobs during execution.
- interconnection scheme**—Design that describes how a multiprocessor system physically connects its components, such as processors and memory modules.
- Interface Definition Language (IDL)**—A language used to specify the details of the RPCs, which provides a language-independent representation of interfaces and allows distributed applications to transparently call procedures on remote computers.
- Interface Definition Language (IDL) file** (Windows XP)—File that specifies the interfaces that an RPC server exposes.
- interface ID (IID)**—Globally unique ID for a COM interface.
- interfault time**—Time between a process's page faults. This is used in the page-fault-frequency page-replacement strategy to determine when to increase or decrease a program's page frame allocation.
- interlocked singly linked list (SList)** (Windows XP)—Singly linked list in which insertions and deletions are performed as atomic operations.
- interlocked variable access** (Windows XP)—Method of accessing variables that ensures atomic reads and writes to shared variables.
- intermediate code generator**—Stage of the compilation process that receives input from the parser and outputs a stream of instructions to the optimizer.

intermediate driver (Windows XP)—Device driver that can be interposed between high- and low-level drivers to filter or process I/O requests for a device.

intermediate-level scheduling—Determining which processes may enter the low-level scheduler to compete for a processor.

internal fragmentation—Phenomenon in fixed-partition multiprogramming systems in which holes occur when the size of a process's memory and data is smaller than the partition in which the process executes.

Internet—Network of communication channels that provides the backbone for telecommunication and the World Wide Web. Each computer on the Internet determines which services it uses and which it makes available to other computers connected to the Internet.

Internet Connection Firewall (ICF)—Windows XP's packet filtering firewall.

Internet Key Exchange (IKE) (IPSec)—Key-exchange protocol used in IPSec to perform key management, which allows secure key exchange.

Internet Protocol (IP)—Primary protocol for directing information over a network. Destinations on the Internet are specified by 32-bit numbers called IP addresses.

Internet Protocol Security (IPSec)—Transport-layer security protocol that provides data privacy, integrity and authentication.

Internet Protocol version 6 (IPv6)—New version of the Internet Protocol that uses 128-bit addresses and specifies three types of addresses: unicast, anycast and multicast.

Internetwork Packet exchange (IPX)—Novell Netware's network protocol designed specifically for LANs.

interoperability—Permits software components to interact among different hardware and software platforms, programming languages and communication protocols.

interpreter—Application that can execute code other than machine code (e.g., high-level-language instructions).

interprocess communication (IPC) manager—Operating system component that governs communication between processes.

interrupt—Hardware signal indicating that an event has occurred. Interrupts cause the processor to invoke a set of software instructions called an interrupt handler.

Interrupt Dispatch Table (IDT)—Kernel data structure that maps hardware interrupts to interrupt vectors.

interrupt handler—Kernel code that is executed in response to an interrupt.

interrupt request level (IRQL) (Windows XP)—Measure of interrupt priority; an interrupt that occurs at an IRQL equal to or lower than the current IRQL is masked.

interrupt service routine (ISR) (Windows XP)—Function, registered by a device driver, that processes interrupts issued by the device that the driver services.

interrupt vector—Array in protected memory containing pointers to the locations of interrupt handlers.

interrupting clock—Hardware device that issues an interrupt after a certain amount of time (called a quantum), e.g., to prevent a process from monopolizing a processor. This can ensure that a processor will not be monopolized by a malicious or malfunctioning process.

interval timer—See interrupting clock.

intrusion detection system (IDS)—Application that monitors networks and application log files, which record information about system behavior, such as the time at which operating services are requested and the name of the process that requests them.

invalidate—To invalidate a file, the client checks the time stamp of the copy of the file on the server. If this copy has been updated since the client last copied the file, then the client must download the latest version. If the server copy has not been updated since the client copied it, then the client can work on its cached copy.

invalidation—Memory-coherence protocol in which a process first invalidates—i.e., voids—all other copies of a page before writing to the page.

inverted page table—Page table containing one entry for each page frame in main memory. Inverted page tables incur less table fragmentation than traditional page tables, which typically maintain in memory a greater number of page table entries than page frames. Hash functions map virtual page numbers to an index in the inverted page table.

IOStone—Popular synthetic benchmark that evaluates file systems.

IP address—Address of a particular host on the Internet.

IP spoofing—Attack in which an attacker simulates the IP address of an authorized user or host to obtain unauthorized access to resources.

J

Jakarta project—Provides commercial-quality server solutions based on the Java platform that are developed in an open and cooperative fashion.

Java—Object-oriented programming language developed by Sun Microsystems that promotes portability by running on a virtual machine.

Java Community Process—Open organization that focuses on developing Java technology specifications, including servlet and JavaServer Pages.

Java Management Extensions (JMX)—Developed by Sun and network management industry leaders, which defines a component framework that enables developers to build automated, intelligent and dynamic network management solutions.

Java Virtual Machine (JVM)—Virtual machine that enables Java programs to execute on many different architectures without recompiling Java programs into the native machine language of the computer on which they execute. The JVM promotes application portability and simplifies programming by freeing the programmer from architecture-specific considerations.

JavaServer Pages (JSP)—Allows Web-page programmers to create pages that use encapsulated Java functionality and even to write scriptlets of actual Java code directly in the page.

JavaSpaces—Jini service that implements a simple, high-level architecture for building distributed systems. The JavaSpaces service provides distributed, shared storage (and shared memory) for Java objects and enables Java objects to communicate, share objects and coordinate tasks using the storage.

Jini—Framework for building reliable and fault-tolerant distributed systems with existing Java technologies. Jini extends the idea of providing services beyond industry-based computer networks and into home-based networks.

job—Set of work to be done by a computer.

job-aware scheduling—Multiprocessor scheduling algorithms that account for job properties when making scheduling decisions; these algorithms typically attempt to maximize parallelism or processor affinity.

job-blind scheduling—Multiprocessor scheduling algorithms that ignore job properties when making scheduling decisions; these algorithms are typically simple to implement.

job control language—Commands interpreted by a job stream processor that define and facilitate the setup of the next job in a single-stream batch-processing system.

job object (Windows XP)—Object that groups several processes and allows developers to manipulate and set limits on these processes as a group.

job scheduling—See high-level scheduling.

job stream processor—Entity in single-stream batch-processing systems that controls the transition between jobs.

job-to-job transition—Time during which jobs cannot execute in single-stream batch-processing systems while one job is purged from the system and the next job is loaded and prepared for execution.

join—Thread operation that causes the calling thread to block until the thread it joins terminates. A primary thread often joins each of threads it creates so that its corresponding process does not exit until all of its threads have terminated.

join (database)—Operation that combines relations.

journaling file system—See log-structured file system (LFS).

JXTA—Project created at Sun Microsystems, Inc., which creates a standard, low-level, platform- and language-independent protocol that promotes interoperability among peer-to-peer applications.

K

Kerberos—Freely available, open-source authentication and access control protocol developed at MIT that provides protection against internal security attacks. It employs secret-key cryptography to authenticate users in a network and to maintain the integrity and privacy of network communications.

kernel—Software that contains the core components of an operating system.

kernel control path (Linux)—A kernel execution context that may perform operations requiring mutual exclusive access to kernel data structures.

kernel handle (Windows XP)—Object handle accessible from any process's address space, but only in kernel mode.

kernel-level thread—Thread created by an operating system (also called kernel thread).

kernel mode—Execution mode of a processor that allows processes to execute privileged instructions.

kernel-mode APC (Windows XP)—APC generated by a kernel-mode thread and queued to a specified user-mode thread; the user-mode thread must process the APC as soon as the user-mode thread obtains the processor.

kernel-mode driver—Device driver that executes in kernel mode.

kernel process (KPROCESS) block (Windows XP)—Kernel data structure that stores information about a process, such as that process's base priority class.

kernel program—Typical program that might be run at an installation; it is executed "on paper" using manufacturers' instruction timings and used for application-specific performance evaluation.

kernel semaphore (Linux) — Semaphore implemented by the kernel to provide mutual exclusion.

kernel thread (Linux)—Thread that executes kernel code.

kernel thread (KTHREAD) block (Windows XP)—Kernel data structure that stores information about a thread, such as the objects on which that thread is waiting and the location in memory of the thread's kernel stack.

key—Input to a cipher to encrypt data; keys are represented by a string of digits.

key agreement protocol—Rules that govern key exchange between two parties over an insecure medium.

key distribution center (KDC) — Central authority that shares a different secret key with every user in the network.

key generation—Creation of encryption keys.

ksoftirqd (Linux)—Daemon that schedules software interrupt handlers when softirq load is high.

kswapd (Linux)—Daemon that swaps pages to disk.

Lamport's bakery algorithm—N-thread mutual exclusion algorithm based on a "take a ticket" system.

lane—Route between two points in a PCI Express bus. PCI Express devices are connected by a link that may contain up to 32 lanes.

large page (Windows XP)—Set of pages contiguous in memory that the VMM treats as a single page.

last processor (Windows XP)—Thread attribute equal to the processor that most recently executed the thread.

latency (process scheduling)—Time a task spends in a system before it is serviced.

laxity—Value determined by subtracting the sum of the current time and a process's remaining execution time from the process's deadline. This value decreases as a process nears its deadline.

- layer**—Level of abstraction in the TCP/IP protocol stack associated with certain conceptual functions. These layers are the application layer, transport layer, network layer and link layer.
- layered biometric verification (LBV)**—Authentication technique that uses multiple measurements of human features (such as face, finger and voice prints) to verify a user's identity.
- layered operating system**—Modular operating system that places similar components in isolated layers. Each layer accesses the services of the layer below and returns results to the layer above.
- lazy allocation**—Policy of waiting to allocate resources, such as pages in virtual memory and page frames in main memory, until absolutely necessary.
- lazy copying**—Process migration strategy that transfers pages from the sender only when the process at the receiving node references these pages.
- lazy data propagation**—Technique in which writing processors send coherence information after a release, but not the data; a processor retrieves the data when it accesses a page that it knows is not coherent.
- lazy migration**—Process migration strategy in multiprocessor systems that does not transfer all pages during initial migration. This increases residual dependency but reduces initial migration time.
- lazy release consistency**—Memory-coherence strategy in which a processor does not send coherence information after writing to a page until a new processor attempts an acquire operation on that memory page.
- lease**—Agreement between the client and server for controlling file locks.
- least-frequently-used (LFU) page replacement**—Page-replacement strategy that replaces the page that is least frequently used or least intensively referenced. LFU is easy to implement, but generally does not predict future page usage well.
- least-recently-used (LRU) page replacement**—Page-replacement strategy that replaces the page that has not been referenced for the longest time. LRU generally predicts future page usage well but incurs significant overhead.
- Lempel-Ziv compression algorithm**—Data compression algorithm that NTFS uses to compress files.
- level (RAID)**—A particular organization of a RAID system, such as level 1 (mirroring) or level 2 (Hamming ECC parity). See also RAID level 0, RAID level 1, RAID level 2, RAID level 3, RAID level 4 and RAID level 5.
- level of multiprogramming**—See degree of multiprogramming.
- lexer**—See lexical analyzer.
- lexical analyzer**—Part of a compiler that separates the source code into tokens.
- library module**—Precompiled module that performs common computer routines, such as I/O routines or mathematical functions.
- Lightweight Directory Access Protocol (LDAP) (Windows XP)**—Protocol for accessing, searching and modifying Internet directories (e.g., Active Directory).
- lightweight process (LWP)**—A single thread of program instructions (also called a thread of execution or thread of control). Threads are "lightweight" because they share their address space with other threads in the same process.
- limit register**—Register used in fixed-partition multiprogramming systems to mark where a process's memory partition ends.
- linear address space (IA-32 Intel architecture)**—32-bit (4GB) virtual address space. Under pure segmentation, this address space is mapped directly to main memory. Under segmentation/paging, this address space is divided into page frames that are mapped to main memory.
- linear ordering (Havender)**—Logical arrangement of resources that requires that processes request resources in a linear order. This method denies the circular-wait necessary condition for deadlock.
- link (file system)**—Directory entry that references an existing file. Hard links reference the location of the file on its storage device; soft links store the file's pathname.
- link (network)**—Medium over which services are physically transmitted in a network.
- link layer (in TCP/IP)**—Responsible for interfacing with and controlling the physical medium over which data is sent.
- linking**—Process of integrating a program's object modules into a single executable file.
- linking loader**—Application that performs both linking and loading.
- Linux security modules (LSM) framework (Linux)**—Framework that allows system administrators to specify the access control mechanism employed by the system.
- Linux Standard Base (Linux)**—Project that aims to specify a standard Linux interface to improve application portability between kernel versions (and distributions).
- list (file)**—Operation that prints or displays a file's contents.
- lmbench microbenchmark**—Microbenchmark suite that enables evaluators to measure and compare system performance on a variety of UNIX platforms.
- load**—See request rate.
- load balancer**—Node in a load balancing cluster that distributes the workload (such as thousands of requests from the clients) to a set of nodes so that all hardware is efficiently utilized.
- load balancing**—Operation that attempts to evenly distribute system load between processors in the system.
- load balancing cluster**—Cluster in which one particular node works as a load balancer to distribute the load to a set of nodes, so that all hardware is efficiently utilized.
- load module**—Integrated module produced by a linker that consists of object code and relative addresses.
- loadable kernel module (Linux)**—Software that can be integrated into the kernel at runtime.
- loader**—Application that loads linked executable modules into memory.
- local area network (LAN)**—Type of network used to interconnect resources using high-speed communication paths optimized for

- local area environments, such as office buildings or college campuses.
- local descriptor table (LDT)** (IA-32 Intel architecture)—Segment map table that contains mapping information for process segments. The system may contain up to 8,191 LDTs, each containing 8,192 entries.
- local procedure call (LPC)** (Windows XP)—Procedure call made by a thread in one process to a procedure exposed by another process in the same domain (i.e., set of computers that share common resources); LPCs can be created only by system components.
- local remote procedure call (LRPCs)** (Windows XP)—RPCs between two processes on the same machine.
- locality**—Empirical phenomenon describing events that are closely related in space or time. When applied to memory access patterns, spatial locality states that when a process references a particular address, it is also likely to access nearby addresses; temporal locality states that when a process references a particular address, it is likely to reference it again soon.
- localized least-recently used page replacement** (Windows XP)—Policy of moving to disk the least-recently used page of a process at its working set maximum when that process requests a page in main memory. The policy is localized by process, because Windows XP moves only the requesting process's pages; Windows XP approximates this policy using the clock algorithm.
- location** (file)—Address of a file on a storage device or in the system's logical file organization.
- location-dependent call**—System call that depends on the workstation (in Sprite) on which the call is executed. Location-dependent calls produce different results for different workstations.
- location-independent call**—System call that does not depend on the workstation (in Sprite) on which the call is executed. Location-independent calls produce the same result for all workstations.
- location transparency**—Builds on access transparency to hide the location of resources in a distributed system from those attempting to access them.
- lockstep synchronization**—Situation where asynchronous threads execute code in strict alternation.
- log file**—Records information about system behavior, such as the time at which operating services are requested and the name of the process that requests them.
- log-structured file system (LFS)**—File system that performs all file operations as transactions to ensure that file system data and metadata is always in a consistent state. An LFS generally exhibits good write performance because data is also appended to the end of a system-wide log file. To improve read performance, an LFS typically distributes metadata throughout the log and employs large caches to store that metadata so that the locations of file data can be found quickly.
- logic bomb**—Virus that executes its code when a specified condition is met.
- logical address space** (IA-32 Intel architecture)—Set of addresses contained in a segment.
- logical backup**—Backup technique that stores file data and the file system's directory structure, often in a common, compressed format.
- logical block**—See logical record.
- logical clock**—Assigns a timestamp to each event that happens in a system in order to create a total ordering of events.
- logical record**—Collection of data treated as a unit by software.
- logical topology**—Map of a network that depicts which nodes are directly connected.
- logical view**—View of files that hides the devices that store them, their format and the system's physical access techniques.
- long-term scheduling**—See high-level scheduling.
- LOOK disk scheduling**—Variation of the SCAN disk scheduling strategy that "looks" ahead to the end of the current sweep to determine the next request to service. If there are no more requests in the current direction, LOOK changes the preferred direction and begins the next sweep, stopping when passing a cylinder that corresponds to a request in the queue. This strategy eliminates unnecessary seek operations experienced by other variations of SCAN by preventing the read/write head from moving to the innermost or outermost cylinder unless it needs to service a request at those cylinders.
- lookup service**—Heart of the Jini architecture, which maintains information about available Jini services and enables clients to discover and use them.
- loopback device** (Linux)—Virtual device that enables operations to be performed on data between layers of a system service (e.g., the file system).
- loosely coupled system**—System in which processors do not share most resources; these systems are flexible and fault tolerant but perform worse than tightly coupled systems.
- low-level driver** (Windows XP)—Device driver that controls a peripheral device and does not depend on any lower-level drivers.
- low-level scheduling**—Determining which process will gain control of a processor.
- Lucifer algorithm**—Encryption algorithm created by Horst Feistel of IBM, which was chosen as the DES by the United States government and the National Security Agency (NSA) in the 1970s.

M

- Mac OS**—Line of operating systems for Apple Macintosh computers first introduced in 1997.
- Mach**—Early microkernel operating system, designed at Carnegie-Mellon University by a team led by Richard Rashid. Mach has influenced the design of Windows NT and has been used to implement Mac OS X.
- machine language**—Language that is defined by a computer's hardware design and can be natively understood by the computer.
- Macintosh**—Apple Computer's PC line that introduced the GUI and mouse to mainstream computer users.

- magnetic tape storage**—Rewritable magnetic storage medium that accesses data sequentially. Its sequential nature makes it unsuitable for direct access applications.
- mailslot** (Windows XP)—Message queue, which a process can employ to receive messages from other processes.
- mailslot client** (Windows XP)—Process that sends mailslot messages to mailslot servers.
- mailslot server** (Windows XP)—Process that creates a mailslot and receives messages in it from mailslot clients.
- main memory** — Volatile memory that stores instructions and data; it is the lowest level of the memory hierarchy that can be directly referenced by a processor.
- main thread of execution**—Thread created upon process creation (also called primary thread).
- mainboard**—Printed circuit board that provides electrical connections between computer components such as processor, memory and peripheral devices.
- major device identification number** (Linux)—Value that uniquely identifies a device in a particular device class. The kernel uses this value to determine a device's driver.
- major function code** (Windows XP)—Field in an IRP that describes the general function (e.g., read or write) that should be performed to fulfill an I/O request.
- major version number** (Linux)—Value that uniquely identifies a significant Linux release.
- manager level** (in JMX)—Level that gives a management application access to managed resources (created in the instrumentation level) and operates these resources via the JMX agents.
- mandatory access control (MAC)**—Access control model in which policies predefine a central permission scheme by which all subjects and objects are controlled.
- manual-reset timer object** (Windows XP)—Timer object that remains *signaled* until a thread specifically resets the timer.
- many-to-many (m-to-n) thread mapping**—Threading model in which a set of user threads is assigned to a set of kernel threads so that applications can benefit both from kernel-level threads and user-level features such as scheduler activations. In practice the number of user threads is greater than or equal to the number of kernel threads in the system to minimize memory consumption.
- many-to-one thread mapping**—Threading model in which all user-level threads in a process are assigned to one kernel thread.
- marshaling of data**—A routine for the client stub to package procedure arguments and return values for transmission over a network.
- mask a signal**—Prevent a signal from being delivered. Signal masking enables a multithreaded process to specify which of its threads will handle signals of a particular type.
- massive parallelism**—Property of a system containing large numbers of processors so that many parts of computations can be performed in parallel.
- massively parallel processor**—Processor that performs a large number of instructions on large data sets at once; array processors are often called massively parallel processors.
- Master File Table (MFT)** (NTFS)—File which is structured as a table in which NTFS stores information (e.g. name, time stamp, and location) about all files in the volume.
- master node** (in a Beowulf cluster)—Node, also known as the head node, that acts as a server to distribute the workload, control the access to the cluster and handle the shared resources.
- master/slave multiprocessor organization**—Scheme for delegating operating system responsibilities in which only one processor (the "master") can execute the operating system, and the other processors (the "slaves") can execute only user processes.
- maximum need** (Dijkstra's banker's algorithm)—Characteristic of a process in Dijkstra's banker's algorithm that describes the largest number of resources (of a particular type) the process will need during execution.
- MD5 Message Digest Algorithm**—Hash algorithm developed by Professor Ronald L. Rivest at MIT that is widely used to implement digital signatures.
- mean** — Average of a set of values.
- mean response time** (disk scheduling)—Average time a system spends waiting for a disk request to be serviced.
- mean-time-to-failure (MTTF)** (RAID)—Average time before a single-disk failure.
- member**—Sequential subfile of a partitioned file.
- memory coherence**—State of a system in which the value obtained from reading a memory address is always the same as the most-recently written value at that address.
- memory compaction**—Relocating all partitions in a variable-partition multiprogramming system to one end of main memory to create the largest possible memory hole.
- memory descriptor list (MDL)** (Windows XP)—Data structure used in an I/O transfer that maps a process's virtual addresses to be accessed in the transfer to physical memory addresses.
- memory dump** (Linux)—Action that generates a core file before terminating a process.
- memory footprint** (Linux)—Size of unswappable memory consumed by the kernel.
- memory hierarchy**—Classification of memory from fastest, lowest-capacity, most expensive memory to slowest, highest-capacity, least expensive memory.
- memory line**—Entry in memory that stores one machine word of data, which is typically four or eight bytes.
- memory management strategy**—Specification of how a particular memory organization performs operations such as fetching, placing and replacing memory.
- Memory Management Unit (MMU)**—Special-purpose hardware that performs virtual-to-physical address translation.

- memory manager**—Component of an operating system that implements the system's memory organization and memory management strategies.
- memory map**—Data structure that stores the correspondence between a process's virtual address space and main memory locations.
- memory-mapped file**—File whose data is mapped to a process's virtual address space, enabling a process to reference file data as it would other data. Memory-mapped files are useful for programs that frequently access file data.
- memory organization**—Manner in which the system views main memory, addressing concerns such as how many processes exist in memory, where to place programs and data in memory and when to replace those pieces with other pieces.
- memory pool** (Linux)—Region of memory reserved by the kernel for a process to ensure that the process's future requests for memory are not denied.
- memory protection**—Mechanism that prevents processes from accessing memory used by other processes or the operating system.
- merge I/O requests** (Linux)—To combine two I/O requests to adjacent locations on disk into a single request.
- mesh network**—Network in which at least two nodes have more than one path connecting them. Faster and more fault tolerant than all but fully-connected mesh networks.
- message**—IPC mechanism that allows data to be transmitted by specifying a message type and variable-length field of data.
- message digest**—Hash value produced by algorithms such as SHA-1 and MD5 when applied to a message.
- message integrity**—Property indicating whether message has been altered in transmission.
- message passing**—Mechanism to allow unrelated processes to communicate by exchanging data.
- message queue** (Linux)—Structure that stores messages that have yet to be delivered to processes.
- message queue descriptor** (Linux)—Structure that stores data regarding a message queue.
- metadata**—Data that a file system uses to manage files and that is inaccessible to users directly. Inodes and superblocks are examples of metadata.
- method** (of an object)—Part of an object that manipulates object attributes or performs a service.
- micro-op**—Simple, RISC-like instruction that is the only type of instruction processed by a Pentium processor; the Pentium's instruction decoder converts complex instructions into a series of micro-ops.
- microbenchmark**—Performance evaluation tool that measures the speed of a single operating system operation (e.g., process creation).
- microbenchmark suite**—Program that consists of a number of microbenchmarks, typically used to evaluate many important operating system operations.
- microcode**—Microprogramming instructions.
- microkernel operating system**—Scalable operating system that puts a minimal number of services in the kernel and requires user-level programs to implement services generally delegated to the kernel in other types of operating systems.
- microprogramming**—Layer of programming below a computer's machine language that includes instructions necessary to implement machine-language operations. This enables processors to divide large, complex instructions into simpler ones that are performed by its execution unit.
- Microsoft IDL (MIDL)**—Microsoft's extension of Interface Definition Language (IDL)—The Open Group's Distributed Computing Environment (DCE) standard for RPC interoperability.
- middleware**—Layer of software that enables communication between different applications. Middleware simplifies application programming by performing work such as network communication and translation between different data formats.
- migration transparency**—Masks the movement of an object from one location to another in the system, such as the movement of a file from one server to another.
- minimum-laxity-first**—Scheduling policy that assigns higher priority to processes that will finish with minimal processor usage.
- minor device identification number** (Linux)—Value that uniquely identifies devices that are assigned the same major number (e.g., a hard drive partition).
- minor function code** (Windows XP)—Field in an IRP that, together with the major function code, describes the specific function that should be performed to fulfill an I/O request (e.g., to start a device, a PnP major function code is used with a start device minor function code).
- minor version number** (Linux)—Value that identifies successive stable (even) and development (odd) versions of the Linux kernel.
- MIPS (million instructions per second)**—Unit commonly used to categorize the performance of a particular computer; a rating of one MIPS means a processor can execute one million instructions per second.
- mirroring** (RAID)—See disk mirroring.
- missing-segment fault**—Fault that occurs when a process references a segment that is not currently in main memory. The operating system responds by loading the segment from secondary storage into main memory when space is available.
- mission-critical system**—System that must function properly; its failure could lead to loss of property, money or even human life.
- mixed model**—COM threading model, in which some COM objects reside in single apartments and others can be accessed by free threads.
- MobileMark**—Popular benchmark for evaluating systems installed on mobile devices developed by Business Application Performance Corporation (BAPCo).
- modified bit**—Page table entry bit that indicates whether a page has been modified and hence must be copied to secondary storage before being replaced (also known as the dirty bit).

- Modified No-write Page List** (Windows XP)—List of page frames for which the VMM must write an entry to a log file before freeing.
- Modified Page List** (Windows XP)—List of page frames that the VMM must write to the pagefile before freeing.
- module**—Independently developed subprogram that can be combined with other subprograms to create a larger, more complex program; programmers often use precompiled library modules to perform common computer functions such as I/O manipulations or random number generation.
- monitor**—Concurrency construct that contains both the data and procedures needed to provide mutual exclusion while allocating a serially reusable shared resource or group of serially reusable shared resources.
- monitor entry routine**—Monitor routine that can be called by any thread, but that can be executed by only one thread at a time. Unlike private monitor routines, which can be called only by threads executing inside the monitor, monitor entry routines enforce mutual exclusion.
- monolithic operating system**—Operating system whose kernel contains every component of the operating system. The kernel typically operates with unrestricted access to the computer system.
- Moore's law**—Prediction regarding the evolution of processor design that asserts the number of transistors in a processor will double approximately every 18 months.
- most trusted user status**—See kernel mode.
- motherboard**—See mainboard.
- mount**—Insert a file system into a local directory structure.
- mount operation**—Operation that combines disparate file systems into a single namespace so they can be accessed by a single root directory.
- mount point**—User-specified directory within the native file system hierarchy where the mount command places the root of a mounted file system.
- mount tables**—Tables that store the locations of mount points and their corresponding devices.
- moving-arm assembly**—See disk arm.
- MS-DOS**—Popular operating system for the first IBM Personal Computer and compatible microcomputers.
- multi-op instruction**—Instruction word used by an EPIC system in which the compiler packages a number of smaller instructions for the processor to execute in parallel.
- multicast**—Type of IPv6 address used to send packets to all hosts in a group of related hosts.
- multicomputer system**—System in which processors do not share a common memory or bus. Each processor has access to its own resources. These independent processors are connected in a network to operate cooperatively to form a multicomputer system.
- Multics**—One of the first operating systems to implement virtual memory. Developed by MIT, GE and Bell Laboratories as the successor to MIT's CTSS.
- Multics Relational Data Store (MRDS)**—First commercial relational database system, included in Multics.
- multilevel feedback queue**—Process scheduling structure that groups processes of the same priority in the same round-robin queue. Processor-bound processes are placed in lower-priority queues because they are typically batch processes that do not require fast response times. I/O-bound processes, which exit the system quickly due to I/O, remain in high-priority queues. These processes often correspond to interactive processes that should experience fast response times.
- multilevel paging system**—Technique that enables the system to store portions of a process's page table in discontinuous locations in main memory and store only those portions that a process is actively using. Multilevel page tables are implemented by creating a hierarchy of page tables, each level containing a table that stores pointers to tables in the level below. The bottom-most level is comprised of tables containing the page-to-page-frame mappings. This reduces memory waste compared to single-level page tables, but incurs greater overhead due to the increased number of memory accesses required to perform address translation when corresponding mappings are not contained in the TLB.
- multiple-instruction-stream, multiple-data-stream (MIMD) computer**—Computer architecture consisting of multiple processing units, which execute independent instructions and manipulate independent data streams; this design describes multiprocessors.
- multiple-instruction-stream, single-data-stream (MISD) computer**—Computer architecture consisting of several processing units, which execute independent instruction streams on a single stream of data; these architectures have no commercial application.
- multiple shared bus architecture**—Interconnection scheme that employs several shared buses connecting processors and memory. This reduces contention, but increases cost compared to a single shared bus.
- Multiple UNC Provider (MUP)** (Windows XP)—File system driver that determines the appropriate redirector to which to send a network I/O request.
- Multiple Virtual Spaces (MVS)**—IBM operating system for System/370 mainframes allowing any number of 16MB virtual address spaces.
- multiprocessing system**—Computing system that employs more than one processor.
- multiprogramming**—Ability to store multiple programs in memory at once so that they can be executed concurrently.
- Multipurpose Internet Mail Extensions (MIME)**—Electronic mail standard defining five content types: text, image, audio, video and application.
- multistage network**—Multiprocessor interconnection scheme that uses switch nodes as hubs for communication between processor nodes that each have their own local memory.
- multithreading**—Technique that incorporates multiple threads of execution within a process to perform parallel activities, possibly simultaneously.
- must-succeed request** (Windows XP)—Request for space in main memory that a process issues when it requires more main memory to continue functioning properly. Windows XP always denies

must-succeed requests; previous versions of Windows always fulfilled them.

mutex object (Windows XP)—Synchronization object that allows at most one thread access to a protected resource at any time; it is essentially a binary semaphore.

mutual exclusion—Restriction whereby execution by a thread of its critical section precludes execution by other threads of their critical sections. Mutual exclusion is crucial to correct execution when multiple threads access shared writable data.

mutual exclusion lock—Variable that indicates if a thread is executing its critical section; if the lock indicates that a thread is in its critical section, other threads are locked out of their own critical sections.

mutual exclusion necessary condition for deadlock—One of the four necessary conditions for deadlock; states that deadlock can occur only if processes cannot claim exclusive use of their resources.

mutual exclusion primitives—Fundamental operations that are needed to implement mutual exclusion: `enterMutualExclusion()` and `exitMutualExclusion()`.

N

n-ary relation—Relation of degree n .

JV-Step SCAN disk scheduling—Disk scheduling strategy that services the first n requests in the queue using the SCAN strategy. When the sweep is complete, the next n requests are serviced. Arriving requests are placed at the end of the request queue. JV-Step SCAN offers good performance due to high throughput, low mean response times and a lower variance of response times than SSTF and SCAN.

n-tier system—Architecture for network-based applications. The three-tier system, for example, has a client tier, an application logic tier and a data tier.

named pipe (Linux)—Pipe that can be accessed via the directory tree, enabling processes that are not parent and child to communicate using pipes. See also pipe.

named pipe (Windows XP)—Type of pipe that can provide bidirectional communication between two processes on local or remote machines and that supports both synchronous and asynchronous communication.

namespace—Set of files that can be identified by a file system.

National Institute of Standards and Technology (NIST)—Organization that sets cryptographic (and other) standards for the U.S. government.

native API (Windows XP)—Programming interface exposed by the executive, which environment subsystems employ to make system calls.

NDIS intermediate driver (Windows XP)—NDIS driver that can be interposed between a miniport driver and a higher-level driver and adds extra functionality, such as translating packets between different communication media, filtering packets or providing load balancing across several NICs.

NDIS miniport driver (Windows XP)—NDIS driver that manages a NIC and sends and receives data to and from the NIC.

NDIS protocol driver (Windows XP)—NDIS driver that places data into packets and passes these packets to lower-level drivers. NDIS protocol drivers provide an interface between the transport drivers and other NDIS drivers and can be used as the lowest layer in the implementation of a transport protocol stack such as TCP/IP.

necessary condition for deadlock—Condition that must be true for deadlock to occur. The four necessary conditions are the mutual exclusion condition, no-preemption condition, wait-for condition and circular-wait condition.

negative feedback—Data informing the system that a resource is having difficulty servicing all requests and the processor should decrease the arrival rate for requests at that resource.

neither I/O (Windows XP)—I/O transfer technique in which a high-level driver executes in the context of the calling thread to set up a buffered I/O transfer or direct I/O transfer or to directly perform the transfer in the process's address space.

.NET—Microsoft initiative aimed at transforming computing from an environment in which users simply execute applications on a single computer to a fully distributed environment; .NET provides a platform- and language-neutral development environment for components that can easily interoperate.

.NET framework—Programming model for creating XML-based Web services and applications. The .NET framework supports over 20 programming languages and facilitates application programming by providing libraries to perform common operations (e.g., input/output, string manipulation and network communication) and access data via multiple database interfaces (e.g., Oracle and SQL Server).

NetBIOS Extended User Interface (NetBEUI) (Windows XP)—Native network and transport protocol for DOS and older Windows operating systems.

NetBIOS over TCP/IP (NBT) (Windows XP)—Replacement protocol for NetBEUI that provides backward compatibility with older applications (by keeping the NetBIOS interface) and takes advantage of TCP/IP protocols.

netfilter framework (Linux)—Mechanism that allows kernel modules to directly inspect and modify packets. This is useful for applications such as firewalls, which modify each packet's source address before the packet is transmitted.

network-based intrusion detection—IDS that monitors traffic on a network for any unusual patterns that might indicate DoS attacks or attempted entry into a network by an unauthorized user.

Network Basic Input/Output System (NetBIOS) (Windows XP)—API used to support NetBEUI and now used with NBT.

Network Data Representation (NDR)—Standard format for network data described in the The Open Group's Distributed Computing Environment (DCE) standard.

network diameter—Shortest path between the two most remote nodes in a system.

Network Driver Interface Specification (NDIS) (Windows XP)—Specification that describes a standard interface between lower-level

layered drivers in the network driver stack; NDIS drivers provide the functionality of the link layer in the TCP/IP stack and are used on Windows XP systems.

network file system—System that allows clients to access files on remote computers. Network file systems do not provide location transparency as distributed file systems do.

Network File System (NFS)—Current de facto standard created by Sun Microsystems for network file sharing, natively supported in most varieties of UNIX (and many other operating systems) with client and server software available for other common platforms.

network layer—Protocols responsible for sending data to the next host toward the destination. This layer exists in both the TCP/IP model and the OSI model of network communication.

network link—Connection between two nodes.

network operating system—Operating system that can manipulate resources at remote locations but does not hide the location of these resources from applications (as distributed systems can).

network redirector—See redirector (Windows XP).

network topology—Representation of the relationships of nodes in a network. Some examples are bus networks, ring networks, star networks, tree networks and mesh networks.

next-fit memory placement strategy—Variation of the first-fit memory placement strategy that begins each search for an available hole at the point where the previous search ended.

nice value (Linux)—Measure of a process's scheduling priority. Processes with a low nice value receive a greater share of processor time than other processes in the system and are therefore "less nice" to other processes in the system.

no-preemption necessary condition for deadlock—One of the four necessary conditions for deadlock; states that deadlock can occur only if resources cannot be forcibly removed from processes.

no-remote-memory-access (NORMA) multiprocessor—Multiprocessor architecture that does not provide global shared memory. Each processor maintains its own local memory. NORMA multiprocessors implement a common shared virtual memory.

node—System component, such as a processor, memory module or switch, attached to a network; sometimes a group of components might be viewed as a single node.

noncontiguous memory allocation—Method of memory allocation that divides a program into several, possibly nonadjacent, pieces that the system places throughout main memory.

nonpaged pool—Area of main memory that stores pages that are never moved to disk.

nonpreemptible resource—Resource that cannot be forcibly removed from a process, e.g., a tape drive. Such resources are the kind that can become involved in deadlock.

nonpreemptive scheduling—Scheduling policy that does not allow the system to remove a processor from a process until that process voluntarily relinquishes its processor or runs to completion.

nonrepudiation—Issue that deals with how to prove that a message was sent or received.

nonresident attribute—NTFS file attribute whose data does not fit inside the MFT entry and is stored elsewhere.

nonuniform-memory-access (NUMA) multiprocessor—Multiprocessor architecture in which each node consists of a processor, cache and memory module. Access to a processor's associated memory module (called local memory) is faster than access to other memory modules in the system.

nonuniform request distribution—Set of disk requests that are not uniformly distributed across disk surfaces. This occurs because processes exhibit spatial locality, leading to localized request patterns.

nonvolatile storage—The contents of nonvolatile storage are not lost when the machine loses power or is powered off.

normal memory (Linux)—Physical memory locations beyond 16MB that the kernel can directly map to its virtual address space. This region is used to store kernel data and user pages.

not-used-recently (NUR) page replacement—Low-overhead approximation to the LRU page-replacement strategy; uses referenced bits and dirty bits to replace a page. NUR first attempts to replace a page that has not been referenced recently and that has not been modified. If no such page is available, the strategy replaces a dirty page that has not been referenced recently, a clean page that has been referenced recently or a referenced page that has been referenced recently, in that order.

notify—Thread operation that transitions its target thread from the *waiting* to the *ready* state.

notify (Java)—Java method that wakes one thread in a monitor's wait set. The thread that is awakened depends on the JVM implementation.

notifyAll—Java method that awakens all threads in a monitor's wait and entry sets. Method `notifyAll` ensures that waiting threads are not indefinitely postponed, but incurs more overhead than `notify`.

O

object—Reusable software component that can model real-world items through properties and actions.

object code—Code generated by a compiler that contains machine-language instructions that must be linked and loaded before execution.

object handle (Windows XP)—Data structure that allows threads to manipulate an object.

Object Linking and Embedding (OLE)—Microsoft technology built on COM that defines a standard method for processes to exchange data by linking or embedding COM objects.

object manager (Windows XP)—Executive component that manages objects.

object manager namespace (Windows XP)—Group of object names in which each name is unique in the group.

object-oriented operating system (OOOS)—Operating system in which components and resources are represented as objects. Object-oriented concepts such as inheritance and interfaces help

- create modular operating systems that are easier to maintain and extend than operating systems built with previous techniques. Many operating systems use objects, but few are written entirely using object-oriented languages.
- object-oriented programming (OOP)**—Style of programming that allows programmers to quickly build complex software systems by reusing components called objects, built from "blueprints" called classes.
- Object Request Broker (ORB)**—Component residing on both CORBA client and server, which is responsible for initiating communication between systems.
- object serialization**—Allows objects to be encoded into byte streams and transmitted from one address space to another.
- offset**—See displacement.
- on-board device**—Device that is physically connected to a computer's mainboard.
- on-demand migration**—Another name for lazy process migration.
- one-to-one mapping**—Threading model in which each user-level thread is assigned to a kernel-level thread.
- online**—State describing a computer that is turned on (i.e., active) and directly connected to a network.
- Online Certificate Status Protocol (OCSP)**—Protocol that validates certificates in real time.
- online spare**—See hot spare disk.
- online transaction processing (OLTP)**—Type of system that typically receives many disk requests to randomly distributed locations containing small amounts of data (e.g., databases and Web servers). Such systems significantly improve performance using disk scheduling algorithms.
- open (file)**—Operation that prepares a file to be referenced.
- Open DataBase Connectivity (ODBC)**—Protocol for middleware that permits applications to access a variety of databases that use different interfaces. The ODBC driver handles connections to the database and retrieves information requested by applications. This frees the application programmer from writing code to specify database-specific commands.
- Open Software Foundation (OSF)**—Coalition of UNIX developers that built the OSF/1 UNIX clone to compete with AT&T's and Sun's Solaris. The OSF and the AT&T/Sun partnership were the participants in the UNIX Wars.
- open-source initiative (OSI)**—Group that supports and promotes open-source software (see www.opensource.com).
- open-source software**—Software that includes the application's source code and is often distributed under the General Public License (GPL) or a similar license. Open-source software is typically developed by teams of independent programmers worldwide.
- OpenBSD**—BSD UNIX system whose primary goal is security.
- operating system**—Software that manages system resources to provide services that allow applications to execute properly. An operating system may manage both hardware and software resources. Operating systems provide an application programming interface to facilitate application development. They also help make system resources conveniently available to users while providing a reliable, secure and responsive environment to applications and users.
- Operationally Critical Threat, Asset and Vulnerability Evaluation (OCTAVE) method**—Technique for evaluating security threats of a system developed at the Software Engineering Institute at Carnegie Mellon University.
- opportunistic lock (oplock)** (Windows XP)—Lock used by a CIFS client to secure exclusive access to a remote file, ensuring that the client can cache data locally and keep its cache coherent.
- optimal (OPT) page replacement**—Unrealizable page-replacement strategy that replaces the page that will not be used until furthest in the future. This strategy has been shown to be optimal.
- optimizer**—Part of the compiler that attempts to improve the execution efficiency and reduce the space requirement of a program.
- Orange Book**—Document published by U.S. Department of Defence (DoD) to establish guidelines for evaluating the security features of operating systems.
- OS/360**—Operating system for the IBM System/360 mainframes. OS/360 had two major options, MFT and MVT, which stood for "Multiprogramming with a Fixed number of Tasks" and "Multiprogramming with a Variable number of Tasks." OS/360-MVT evolved into MVS, the ancestor of the current IBM mainframe operating system z/OS.
- OSF/1**—UNIX clone built by the Open Software Foundation to compete with Solaris.
- out-of-order execution (OOO)**—Technique in which a processor analyzes a stream of instructions and dynamically reorders instructions to isolate groups of independent instructions for parallel execution.
- overlapped I/O** (Windows XP)—Microsoft synonym for asynchronous I/O.
- overlay**—Concept created to enable programs larger than main memory to run. Programs are broken into pieces that do not need to exist simultaneously in memory. An overlay contains one such piece of a program.
- owner** (file access control)—User who created the file.
- ## P
- P operation**—Operation on a semaphore. If the variable in the semaphore is 0, then the P operation blocks the calling thread. If the variable is greater than 0, the operation will decrement the variable by one and allow the calling thread to proceed.
- package** (Linux)—Portion of a distribution containing an application or service. Users can customize their Linux systems by adding and removing packages.
- packet-filtering firewall**—Hardware or software that examines all data sent from outside its LAN and rejects data packets based on predefined rules, such as reject packets that have local network addresses or reject packets from certain addresses or ports.

- page**—Fixed-size set of contiguous addresses in a process's virtual address space that is managed as one unit. A page contains portions of a process's data and/or instructions and can be placed in any available page frame in main memory.
- page cache** (Linux)—Cache storing pages of data from disk. When a process requests data from disk, the kernel first determines if it exists in the page cache, which can eliminate an expensive disk I/O operation.
- page directory entry** (IA-32 Intel architecture)—Entry in a page directory that maps to the base address of a page table, which stores page table entries.
- page directory register**—Hardware register that stores a pointer to the current process's page directory table.
- page directory table** (Windows XP)—4KB page that contains 1,024 entries that point to frames in memory.
- page fault**—Fault that occurs as the result of an error when a process attempts to access a nonresident page, in which case the operating system can load it from disk.
- page-fault-frequency (PFF) page replacement**—Algorithm that adjusts a process's resident page set based on the frequency with which the process is faulting. If a process is switching to a larger working set, then it will fault frequently, and PFF will allocate more page frames. Once the process has accumulated its new working set, the page fault rate will stabilize, and PFF will either maintain the resident page set or reduce it. The key to the proper and efficient operation of PFF is maintaining the thresholds at appropriate values.
- page frame**—Block of main memory that can store a virtual page. In systems with a single page size, any page can be placed in any available page frame.
- page frame database** (Windows XP)—Array that contains the state of each page frame in main memory.
- page global directory**—In a two-tiered multilevel page table, the page global directory is a table of pointers to portions of a process's page table. Page global directories are the top level of a multilevel page table hierarchy.
- page global directory** (Linux)—Virtual memory structure that stores addresses of second-level page-mapping tables.
- page list** (Windows XP)—List of page frames that are in the same state. The eight state lists are: the Valid Page List, the Standby Page List, the Modified Page List, the Modified No-write Page List, the Transitional Page List, the Free Page List, the Zeroed Page List, and the Bad Page List.
- page map table**—See page table.
- page middle directory** (Linux)—Virtual memory structure that stores addresses of third-level page-mapping tables (also called page tables).
- page migration**—Technique in which the system transfers pages to the processor (or processors when used with page replication) that accesses the pages most.
- page-replacement strategy**—Strategy that determines which page to replace to provide space for an incoming page. Page replacement strategies attempt to optimize performance by predicting future page usage.
- page replication**—Technique in which the system maintains multiple copies of a page at different nodes so that it can be accessed quickly by multiple processors.
- page table**—Table that stores entries that map page numbers to page frames. A page table contains an entry for each of a process's virtual pages.
- page table entry (PTE)**—Entry in a page table that maps a virtual page number to a page frame number. Page table entries store other information about a page, such as how the page may be accessed and whether the page is resident.
- page table origin register**—Register that holds the location of a process's page table in main memory; having this information accessible in a high-speed register facilitates rapid virtual-to-physical address translation.
- page trimming** (Windows XP)—Technique in which Windows XP takes a page belonging to a process and sets the page's PTE to invalid to determine whether the process actually needs the page. If the process does not request the page within a certain time period, the system removes it from main memory.
- paged pool** (Windows XP)—Pages in memory that may be moved to the pagefile on disk.
- pagefile** (Windows XP)—File on disk that stores all pages that are mapped to a process's virtual address space but do not currently reside in main memory.
- paging**—Virtual memory organization technique that divides an address space into fixed-size blocks of contiguous addresses. When applied to a process's virtual address space, the blocks are called pages, which store process data and instructions. When applied to main memory, the blocks are called page frames. A page is stored on secondary storage and loaded into a page frame if one is available. Paging trivializes the memory placement decision and does not incur external fragmentation (for systems that contain a single page size); paging does incur internal fragmentation.
- parallel port**—Interface to a parallel I/O device such as a printer.
- parent directory**—In hierarchically structured file systems, the directory that points to the current directory.
- parent process**—Process that has spawned one or more child processes. In UNIX, this is accomplished by issuing a fork system call.
- parity**—Technique that detects an even number of errors in data transmission. Parity information is generated by determining whether the data contains an even (or odd) number of 1's (or 0's). This parity information is generated after transmission and compared to the value generated before transmission. Error-correction codes (ECCs), such as Hamming or XOR, use the parity of a string of bits to detect and correct errors. Parity enables RAID systems to provide fault tolerance with lower storage overhead than mirrored systems.
- parity logging** (RAID)—Technique that increases write performance in RAID systems using parity by postponing writes to the parity disk while the system is busy. Because parity logging stores infor-

mation in volatile memory, parity information could be lost if the system loses power.

parser—Part of the compiler that receives a stream of tokens from the lexical analyzer and groups the tokens so they can be processed by the intermediate code generator.

partial ordering—An ordering of events that follows the happens-before relation. Some events cannot be ordered using this system which is why it is only a partial ordering.

partition (file system)—Area of a disk whose boundaries cannot be crossed by file data. Partitions can reduce disk fragmentation.

partition (real memory organization)—Portion of main memory allocated to a process in fixed- and variable-partition multiprogramming. Programs are placed into partitions so that the operating system can protect itself from user processes and so that processes are protected from each other.

partitioned file—File composed of sequential subfiles.

Pascal—Structured programming language developed in 1971 by Wirth that became popular for teaching introductory programming courses.

passive IRQL (Windows XP)—IRQL at which user- and kernel-mode normally execute and no interrupts are masked. Passive IRQL is the lowest-priority IRQL.

password aging—Technique that attempts to improve security by requiring users to change their passwords periodically.

password protection—Authentication technique that relies on a user's presenting a username and corresponding password to gain access to a resource or system.

password salting—Technique that inserts characters at various positions in the password before encryption to reduce vulnerability to brute-force attacks.

pathname—String identifying a file or directory by its logical name, separating directories using a delimiter (e.g., "/" or "\"). An absolute pathname specifies the location of a file or directory starting at the root directory; a relative pathname specifies the location of a file or directory beginning at the current working directory.

payload—Code inside a logic bomb that is executed when a specified condition is met.

peer—Single computer in a peer-to-peer system.

peer discovery—Finding peers in a peer-to-peer application.

peer group—Logical construct that represents a set of peers. A peer group is one of the basic types of entities in a network built with the JXTA protocols.

peer-to-peer (P2P) application—Distributes processing responsibilities and information to many computers, thus reclaiming otherwise wasted computing power and storage space and eliminating central points of failure. In a peer-to-peer system, each peer performs both client and server functions.

pending signal—Signal that has not been delivered to a thread because the thread is not *running* and/or because the thread has masked signals of that type.

per-node run queue—Process scheduling queue associated with a group of processors; processes entering the queue are scheduled on the associated node's processors independently of the scheduling decisions made in rest of the system.

per-processor run queue—Process scheduling queue associated with a specific processor; processes entering the queue are scheduled on the associated processor independently of the scheduling decisions made in rest of the system.

performance monitoring—Collection and analysis of system performance information for existing systems; the information includes a system's throughput, response times, predictability, bottlenecks, etc.

performance projection—Estimate of the performance of a system that does not exist, useful for deciding whether to build that system or to modify an existing system's design.

periodic real-time process—Real-time process that performs computation at a regular time interval.

periodic timer (Windows XP)—Waitable timer object that reactivates after a specified interval.

Peripheral Components Interconnect (PCI) bus—Popular bus used to connect peripheral devices, such as network and sound cards, to the rest of the system. PCI provides a 32-bit or 64-bit bus interface and supports transfer rates of up to 533MB per second.

persistence transparency—Hides the information about where the resource is stored—memory or disk.

persistent storage—See secondary storage.

phantom deadlock—Situation due to communications delay associated with distributed computing, when a deadlock detection algorithm (DDA) might detect a deadlock that does not exist.

physical address—See real address.

Physical Address Extension (PAE) (IA-32 Intel architecture)—Mechanism that enables IA-32 processors to address up to 64GB of main memory.

physical address space—Range of physical addresses corresponding to the size of main memory in a given computer. The physical address space may be (and is often) smaller than each process's virtual address space.

physical backup—Copy of each bit of the storage device; no attempt is made to interpret the contents of its file system.

physical block—See physical record.

physical device name—Name given to a file that is specific to a particular device.

physical layer (in OSI)—Transmits bits over physical media, such as cables. The data link layer and physical layer in OSI correspond to the link layer in TCP/IP.

physical memory—See main memory.

physical record—Unit of information actually read from or written to disk.

physical view—View of file data concerned with the particular devices on which data is stored, the form the data takes on those devices,

and the physical means of transferring data to and from those devices.

PID (process identifier)—Integer that uniquely identifies a process.

pipe—Interprocess communication mechanism that uses a page of memory as a first-in-first-out buffer to transfer information between processes.

pipe (in JXTA)—Virtual communication channel that connects two or more peers for sending and receiving messages among peers.

pipe buffer (Linux)—Page of data that is used to buffer data written to a pipe.

pipe client (Windows XP)—Process that connects to an existing pipe to communicate with that pipe's server.

pipe server (Windows XP)—Process that creates a pipe and communicates with pipe clients that connect to the pipe.

placement strategy (main memory)—Strategy that determines where in the main memory to place incoming programs and data.

plaintext—Unencrypted data.

Platform for Privacy Preferences (P3P)—Protects the privacy of information submitted to single sign-on and other applications by allowing users to control the personal information that sites collect.

platter—Magnetic disk medium that stores bits on its surfaces.

plug-and-play—Technology that facilitates driver installation and hardware configuration performed by the operating system.

Plug and Play (PnP) manager (Windows XP)—Executive component (which also exists partly in user space) that dynamically recognizes when new devices are added to the system (as long as these devices support PnP), allocates and deallocates resources to devices and interacts with device setup programs.

pluggable authentication module (PAM) (Linux)—Module that can be installed at runtime to incorporate enhanced authentication techniques in the Linux system.

PnP I/O requests (Windows XP)—IRPs generated by the PnP manager to query a driver for information about a device, to assign resources to a device or to direct a device to perform some action.

policy creation authority—Organization that sets policies for obtaining digital certificates.

polling—Technique to discover hardware status by repeatedly testing each device. Polling can be implemented in lieu of interrupts but typically reduces performance due to increased overhead.

polymorphic virus—Virus that attempts to evade known virus lists by modifying its code (e.g., via encryption, substitution, insertion, and the like) as it spreads.

port (hardware interface)—Bus that connects two devices.

port (networking)—Identifies the specific socket on a machine to which to send data. For example, HTTP communicates by default on port 80.

port mapping (Windows XP)—Process of explicitly allowing the Internet Connection Firewall (ICF) to accept all incoming packets to a particular port.

port of Linux—Version of the Linux kernel that is modified to support execution in a different environment.

portability—Property of software that can run on different platforms.

portable operating system—Operating system that is designed to operate on many hardware configurations.

Portable Operating Systems Interface (POSIX)—API based on early UNIX operating systems.

positioning time—Access time plus latency. Positioning time is used by the SPTF strategy to order requests.

positive feedback—Data informing the system that a resource has excess capacity, so the processor can increase the arrival rate for requests at that resource.

power IRQ (Windows XP)—IRQ at which power failure interrupts execute and at which all interrupts except high-IRQ interrupts are masked.

power manager (Windows XP)—Executive component that administers the operating system's power management policy.

power policy (Windows XP)—Policy of a system with regard to balancing power consumption and responsiveness of devices.

precopy—Process migration strategy in which the sender begins transferring dirty pages before the original process is suspended; once the number of untransferred dirty pages at the sender reaches some threshold, the process migrates.

predicate—Logical decision made on a subject (e.g., a branch comparison).

predictability—Measure of the variance of an entity, such as response time. Predictability is particularly important for interactive systems, where users expect predictable (and short) response times.

preemptible resource—Resource that may be removed from a process such as a processor or memory. Such resources cannot be involved in deadlock.

preemption lock counter (Linux)—Integer that is used to determine whether code executing in kernel mode may be preempted. The value of the counter is incremented each time a kernel control path enters a critical section during which it cannot be preempted.

preemptive scheduling—Scheduling policy that allows the system to remove a processor from a process.

preferred direction—Direction in which the disk head is moving in SCAN-based scheduling algorithms.

preferred server (PS) (in Coda)—Member of the AVSG that provides copies of files for Venus.

prefetching—See anticipatory paging.

prefix table (in Sprite)—Table that stores domain information to aid in file lookups. Each entry in the table represents a separate domain and consists of the absolute path to the root directory within the domain, the server which houses the domain and a token that identifies the domain.

prepaging—See anticipatory paging.

- presentation layer** (in OSI)—Solves compatibility problems by translating the application data into a standard format that can be understood by other layers.
- Pretty Good Privacy (PGP)**—Public-key encryption system primarily used to encrypt e-mail messages and files, designed in 1991 by Phillip Zimmermann.
- primary key**—In a relational database, a combination of attributes whose value uniquely identifies a tuple.
- primary thread**—Thread created upon process creation (also called main thread of execution). When the primary thread returns, its process terminates.
- primary thread** (Windows XP)—Thread created when a process is created.
- principle of least privilege**—Resource access policy that states that a user should only be granted the amount of privilege and access that the user needs to accomplish its designated task.
- printed circuit board (PCB)**—Piece of hardware that provides electrical connections to devices that can be placed at various locations throughout the board.
- priority**—Measure of a process's or thread's importance used to determine the order and duration of execution.
- priority array** (Linux)—Structure within a run queue that stores processes of the same priority.
- priority inversion**—Situation which occurs when a high-priority thread is waiting for a resource held by a low-priority thread, and the low-priority thread cannot obtain the processor because of a medium-priority thread; hence, the high-priority thread is blocked from execution by the medium-priority thread.
- priority of a process**—Importance or urgency of a process relative to other processes.
- privacy** (secure transaction)—One of the five fundamental requirements for a successful, secure transaction. Privacy deals with how to ensure that the information transmitted over the Internet has not been captured or passed to a third party without user knowledge.
- private key**—Key in public-key cryptography that should be known only by its owner. If its corresponding public key encrypts a message, only the private key should be able to decrypt it.
- privilege** (access right)—The manner in which a subject can access an object.
- privileged instruction**—Instruction that can be executed only from kernel mode. Privileged instructions perform operations that access protected hardware and software resources (e.g., switching the processor between processes or issuing a command to a hard disk).
- problem state**—See user mode.
- procedural programming language**—Programming language that is based on functions rather than objects.
- process**—Entity that represents a program in execution.
- process cloning**—Creates a copy of a process on a remote machine.
- process control block (PCB)**—Data structure containing information that characterizes a process (e.g., PID, address space and state); also called a process descriptor.
- process descriptor**—See process control block (PCB).
- process environment block (PEB)** (Windows XP)—User-space data structure that stores information about a process, such as a list of DLLs linked to the process and information about the process's heap.
- process identification number (PID)**—Value that uniquely identifies a process.
- process migration**—Transferring a process and its associated state between two processors.
- process priority**—Value that determines the importance of a process relative to other processes. It is often used to determine how a process should be scheduled for execution on a processor relative to other processes.
- process scheduler**—Operating system component that determines which process can gain access to a processor and for how long.
- process state**—Status of a process (e.g., *running*, *ready*, *blocked*, etc.).
- process table**—Table of known processes. In a segmentation/paging system, each entry points to a process's virtual address space, among other items.
- processor**—Hardware component that executes machine-language instructions and enforces protection for system resources such as main memory.
- processor affinity**—Relationship of a process to a particular processor and a corresponding memory bank.
- processor-bound**—Process (or job) that consumes its quantum when executing. These processes (or jobs) tend to be calculation intensive and issue few, if any, I/O requests.
- processor pool**—Component in Amoeba system, which contains a collection of processors, each having its own memory and Ethernet connection.
- processor scheduling discipline**—See processor scheduling policy.
- processor scheduling policy**—Strategy used by a system to determine when and for how long to assign processors to processes.
- proefs (proc file system)** (Linux)—File system built directly into the kernel that provides real-time information about the status of the kernel and processes, such as memory utilization and system execution time.
- producer**—Thread or process that creates and places data into a shared object.
- producer thread**—Thread that creates and places data into a shared object.
- producer/consumer relationship**—Interaction between threads that produce data (called producers) and threads that consume produced data (called consumers) that illustrates many of the intricacies of asynchronous concurrent execution.
- production program**—Program that is run regularly at an installation.
- profile**—Record of kernel activity taken during a real session, which indicates the operating system functions that are used most often and should therefore be optimized.

profile IRQL (Windows XP)—IRQL at which debugger interrupts execute and at which APC, DPC/dispatch, DIRQL and incoming debugger interrupts are masked.

program counter—Pointer to the instruction a processor is executing for a running process. After the processor completes the instruction, the program counter is adjusted to point to the next instruction the processor should execute.

programmed I/O (PIO)—Implementation of I/O for devices that do not support interrupts in which the transfer of every word from memory must be supervised by the processor.

projection (database)—Operation that creates a subset of attributes.

property (of an object)—Part of an object that stores data about the object.

protected variable (semaphores)—Integer variable storing the state of a semaphore that can be accessed and altered only by calling *P* or *V* on that semaphore.

protection—Mechanism that implements a system's security policy by preventing applications from accessing resources and services without authorization.

protection domain—Collection of access rights. Each access right in a protection domain is represented as an ordered pair with fields for the object name and applicable privileges.

protocol—Set of rules that govern how two entities should interact. Common examples include Transmission Control Protocol (TCP), Internet Protocol (IP) and Hypertext Transfer Protocol (HTTP).

protocol sequence (Windows XP)—String used by a client in an RPC call that specifies the RPC protocol, transport protocol and network protocol for that RPC.

Prototype Page Table Entry (PPTE) (Windows XP)—32-bit record that points to a frame in memory that contains either a copy-on-write page or a page that is part of a process's view of a mapped file.

proxy—In DCOM, the client-side stub that is responsible for marshaling and unmarshaling messages.

Pthread (POSIX 1003.1c thread)—Thread that conforms to the POSIX 1003.1c standard.

public (file access control)—File that may be accessed by any member of the system's user community.

public key—Key in public cryptography that is available to all users that wish to communicate with its owner. If the public key encrypts a message, only the corresponding private key can decrypt it.

public-key cryptography—Asymmetric cryptography technique that employs two inversely related keys: a public key and a private key. To transmit a message securely, the sender uses the receiver's public key to encrypt the message. The receiver then decrypts the message using his or her unique private key.

Public Key Infrastructure (PKI)—Technique that integrates public-key cryptography with digital certificates and certificate authorities to authenticate parties in a transaction.

purchase priority—To pay to receive higher priority in a system.

pure paging—Memory organization technique that employs paging only, not segmentation.

pure peer-to-peer application—Also called a decentralized peer-to-peer application. It does not have a server and therefore does not suffer from the same deficiencies as applications that depend on servers.

Q

quad pumping—Technique for increasing processor performance by performing four memory transfers per clock cycle.

quantum—Unit of time during which a process can execute before it is removed from the processor. Helps prevent processes from monopolizing processors.

query language—Language that allows users to search a database for data that meets certain criteria.

queued access method—File access method that does not immediately service user I/O demands. This can improve performance when the sequence in which records are to be processed can be anticipated and requests can be ordered to minimize access times.

queued spin lock (Windows XP)—Spin lock in which a thread releasing the lock notifies the next thread in the queue of waiting threads.

R

race condition—Occurs when multiple threads simultaneously compete for the same serially reusable resource, and that resource is allocated to these threads in an indeterminate order. This can cause subtle program errors when the order in which threads access a resource is important.

RAID (Redundant Array of Independent Disks)—Family of techniques that use an array of disks to improve disk transfer rates while providing fault tolerance.

RAID controller—Special-purpose hardware that efficiently performs operations such as dividing files into strips, forming files from strips, determining the locations of strips in the array and implementing the array's fault-tolerance mechanism.

RAID level 0—RAID system that uses a striped disk array with no redundancy. Level 0 arrays are not fault tolerant; if one disk fails, all the data in the array that depend on the failed disk are lost. Depending on the array's strip size, all data stored in the array could become unusable with the loss of a single disk. Although RAID 0 is not fault tolerant, it is simple to implement, yields high transfer rates and does not incur any storage overhead. Level 0 arrays are implemented in systems where performance is more important than fault tolerance, such as supercomputers.

RAID level 1—RAID system that employs disk mirroring (also called shadowing) to provide redundancy, so that each disk in the array is duplicated. Stripes are not implemented in level 1, reducing both hardware complexity and system performance. While this results in the highest degree of fault tolerance of any RAID level, only half of the array's capacity can be used to store unique data, which increases cost. Level 1 arrays are implemented in systems where high availability is more important than cost, such as database systems.

RAID level 2—RAID system that is striped at the bit level (i.e., each strip stores one bit). Level 2 arrays are designed to reduce the storage overhead incurred by implementing fault tolerance using mirroring. Instead of maintaining redundant copies of each data item, RAID level 2 uses a version of Hamming error-correcting codes (Hamming ECCs) to store parity information that allows the system to detect up to two errors, correct up to one error and determine the location of the error in a stripe. The size of Hamming ECC codes, and thus the number of parity disks, increases according to the logarithm (base 2) of the number of data disks. Thus, level 2 arrays containing a large number of disks incur significantly less storage overhead than level 1 arrays.

RAID level 3—RAID system that stripes data at the bit or byte level. RAID 3 uses XOR (exclusive-or) error-correcting codes (XOR ECCs), which use the logical XOR operation to generate parity information. XOR ECC uses only one disk to hold parity information, regardless of the size of the array. The system can use the parity bits to recover from any single disk failure. Due to parity checking, RAID level 3 reads and writes require access to the entire array. Similar to RAID level 2, this yields high transfer rates when reading and writing large files, but only one request can be serviced at a time.

RAID level 4—RAID system that is striped using fixed-size blocks (typically larger than a byte) and uses XOR ECC to generate parity data that is stored on a single parity disk. Because level 4 arrays enable coarse-grained striping, the system can potentially service multiple read requests simultaneously, if parity is not determined for each read. When servicing a write request, however, the system must update parity information to ensure that no data is lost in the event of a disk failure. This means that write requests must be performed one at a time, creating a write bottleneck.

RAID level 5—RAID system that is striped at the block level and uses XOR ECC for parity, but distributes parity blocks throughout the array of disks. Because parity blocks are distributed across many disks, multiple parity strips can be accessed simultaneously, removing the write bottleneck for many requests. Although RAID level 5 increases performance relative to RAID levels 2-4, it is complex to implement and more costly. Level 5 arrays are considered general-purpose arrays and are often found in file and application servers, enterprise resource planning (ERP) and other business systems.

RAMAC (Random Access Method of Accounting and Control)—First commercial hard drive produced by IBM.

ramfs (Linux)—Region of main memory treated as a block device. The ramfs file system must be formatted before use.

random access memory (RAM)—Memory whose contents can be accessed in any order.

random delay—Interval of time calculated by the exponential backoff method of CSMA/CD before a transceiver can retransmit a frame after a collision.

random (RAND) page replacement—Page-replacement strategy in which each page in main memory has an equal likelihood of being selected for replacement. Although this strategy is fair and incurs little overhead, it does not attempt to predict future page usage.

random policy—Dynamic load balancing policy in which the system arbitrarily chooses a processor to receive a migrated process.

random-scanning algorithm—Algorithm that uses pseudorandom numbers to generate a broad distribution of IP addresses.

random seek pattern—Series of requests to cylinders randomly distributed across disk surfaces. FCFS causes random seek patterns that result in high response times and low throughput.

random variable—Variable that can assume a certain range of values, where each value has an associated probability.

rate-monotonic (RM) scheduling—Real-time scheduling policy that sets priority to a value that is proportional to the rate at which the process must be dispatched.

read (file)—Operation that inputs a data item from a file to a process.

read access (virtual memory)—Access right that enables a process to read data from a page or segment; see also execute access, write access and append access.

read access (file)—Permission to access a file for reading.

read-modify-write cycle (RAID)—Operation that reads a stripe, modifies its contents and parity, then writes the stripe to the array. It is performed for each write request in RAID systems that use parity. Some systems reduce the cost of read-modify-write cycles by caching strips or by updating parity information only periodically.

read-modify-write (RMW) memory operation—Operation that atomically reads the contents of a variable, changes the contents (possibly based on what it has read) and writes the new value to memory. These operations simplify mutual exclusion algorithms by providing atomic operations.

read operation (in JavaSpaces)—Operation that is similar to the take operation, but does not remove the object from the JavaSpaces service so that other clients can still find it.

read/write head—Moving-head disk component that hovers over a disk surface, reading and writing bits as the disk moves.

read/write lock—Lock that allows a single writer process or multiple reading processes (i.e., processes that will not alter shared variables) to enter a critical section.

reader/writer lock (Linux)—Lock that allows multiple threads to concurrently hold a lock when reading from a resource, but only one thread to hold a lock when writing to that resource.

ready (or runnable) state—Thread state from which a thread can transition to the *running* state and execute on a processor. In Windows XP, a *ready* thread transitions to the *standby* state, from which it transitions to the *running* state.

ready list—Kernel data structure that organizes all *ready* processes in the system. The ready list is typically ordered by process scheduling priority.

ready state—Process (or thread) state from which a process (or thread) may be dispatched to the processor.

real address—Address in main memory.

real memory—See main memory.

- real-time priority class** (Windows XP)—Base priority class encompassing the upper 16 priority levels; threads of this class have static priorities.
- real-time scanner**—Software that resides in memory and actively prevents viruses.
- real-time scheduling**—Scheduling policy that bases priority on timing constraints.
- real-time signal** (Linux)—Signal implementation that helps to implement a real-time system by ensuring that no signals are dropped.
- real-time system**—System that attempts to service requests within a specified (usually short) time period. In mission-critical real-time systems (e.g., air traffic control and petroleum refinery monitors), money, property or even human life could be lost if requests are not serviced on time.
- receiver-initiated policy**—Dynamic load balancing policy in which processors with low utilization attempt to find overloaded processors from which to receive a migrated process.
- record**—In the data hierarchy, a group of fields (e.g., for storing several related fields containing information about a student or a customer).
- recovery**—Restoration of the system's data after a failure.
- recovery key** (NTFS)—Key that NTFS stores that can decrypt an encrypted file; administrators can use recovery keys when a user has forgotten the private key needed to decrypt the file.
- redirector** (Windows XP)—File system driver that interacts with a remote server driver to facilitate network I/O operations.
- reduced instruction set computing (RISC)**—Processor-design philosophy that emphasizes small, simple instruction sets and optimization of the most-frequently used instructions.
- redundancy**—Technique that maintains multiple identical resources to enable recovery upon failure.
- Redundant Array of Independent Disks (RAID)**—See RAID.
- reentrant code**—Code that cannot be changed while in use and therefore can be shared among processes and threads.
- referenced bit**—Page table entry bit indicating whether a page has been referenced recently. Several strategies reset this bit to more accurately determine how recently a page has been referenced.
- register**—High-speed memory located on a processor that holds data for immediate use by the processor.
- registry** (Windows XP)—Central database in which user, system and application configuration information is stored.
- reintegration stage** (in Coda)—Stage right after the client reconnects to the system during which Venus asynchronously updates the server using the CML.
- relation**—A set of tuples in the relational model.
- relational model**—A model of data proposed by Codd that is the basis for most modern database systems.
- relative address**—Address that is specified based on its location in relation to the beginning of a module.
- relative path**—Path that specifies the location of a file relative to the current working directory.
- relaxed consistency**—Category of memory-coherence strategies that permit the system to be in an incoherent state for a few seconds after a write, but improve performance over strict consistency.
- release consistency**—Memory-coherence strategy in which multiple accesses to shared memory are considered a single access; these accesses begin with an acquire and end with a release, after which coherence is enforced throughout the system.
- release operation**—In several coherence strategies, this operation indicates that a process is done accessing shared memory.
- reliability**—Measure of fault tolerance. The more reliable a resource, the less likely it is to fail.
- reliable network**—Network which does not damage or lose packets.
- relocatable loading**—Method of loading that translates relative addresses in a load module to absolute addresses based on the location of a requested block of memory.
- relocating**—Process of adjusting the addresses of program code and data.
- relocation transparency**—Masks the relocation of an object from other objects that communicate with it.
- Remote Access Service (RAS)** (Windows XP)—Network service, which allows users to remotely connect to a LAN.
- remote method invocation (RMI)**—Allows Java programmers to implement distributed systems without having to explicitly program sockets.
- remote procedure call (RPC)**—Allows a process executing on one computer to invoke a procedure (or function) in a process executing on another computer.
- remote reference layer (RRL)**—Works with the transport layer to send marshaled messages between the client and server in RMI.
- Remote Spooling Communications Subsystem (RSCS)** (VM)—Component of VM that provides the capability to send and receive files in a distributed system.
- rename (file)**—Operation that changes a file's name.
- reparse point** (NTFS)—File attribute containing a tag and a block of up to 16KB of data that a user associates with a file or directory: when an application accesses a reparse point, the system executes the file system filter driver specified by the reparse point's tag.
- replacement strategy** (main memory)—Method that a system uses to determine which piece of program or data to displace to accommodate incoming programs or data.
- replication**—Provides multiple resources that perform the same function in a system.
- replication transparency**—Hides the fact that multiple copies of a resource are available in the system. All access to a group of replicated resources occurs as if there were one such resource available.
- request IRQ** (Windows XP)—IRQ at which interprocess interrupts execute and all interrupts except power-level and high-level interrupts are masked.

- request list** (Linux)—Structure that stores pending I/O requests. This list is sorted to improve throughput by reducing seek times.
- request rate**—Measure of request frequency. The higher the request rate, the greater the system load. Systems experiencing a high disk request rate tend to benefit from disk scheduling.
- Request to Send (RTS)**—Message sent from a wireless device in the CSMA/CA protocol that indicates a desire to transmit data, the length of the transmission, the sender address and the receiver address. If the medium is available, the receiver will send a Clear To Send (CTS) message.
- reserve memory** (Windows XP)—To indicate to the VMM that a process intends to use an area of virtual memory; the VMM allocates memory in the process's virtual address space but does not allocate any page frames in main memory.
- resident attribute** (NTFS)—File attribute whose data is stored within the MFT entry.
- resident page set**—Set of a process's pages that are currently in memory; these pages may be referenced without generating a page fault. The resident page set might differ in size from a process's working set, which is the set of pages that must be in memory for a process to execute efficiently.
- resident virus**—Virus that, once loaded into memory, executes until the computer is powered down.
- residual dependency**—Dependency of a migrated process on its original node after process migration because some of the process's state remains on the original node.
- resource-allocation graph**—Graph that shows processes and resources in a system. An arrow pointing from a process to a resource indicates that the process is requesting the resource. An arrow pointing from a resource to a process indicates that the resource is allocated to the process. Such a graph helps determine if a deadlock exists and, if so, helps identify the processes and resources involved in the deadlock.
- resource type**—Grouping of resources that perform a common task.
- resources layer** (in grid computing)—Layer that enables applications to query and share a resource.
- response time**—In an interactive system, the time from when a user presses an *Enter* key or clicks a mouse until the system delivers a final response.
- restricted algorithm**—Algorithm that provides security by relying on the sender and receiver to use the same encryption algorithm and maintain its secrecy.
- resume**—Remove a process from a suspended state.
- reverse mapping** (Linux)—Linked list of page table entries that reference a page of memory. This facilitates updating all PTEs corresponding to a shared page that is about to be replaced.
- Rijndael**—Block cipher developed by Dr. Joan Daemen and Dr. Vincent Rijmen of Belgium. The algorithm can be implemented on a variety of processors.
- ring network**—Network consisting of a set of nodes, each maintaining exactly two connections to other nodes in that network. These networks have a low fault tolerance since the failure of any single node can cause the whole network to fail.
- robust operating system**—Operating system that is fault tolerant and reliable—the system will not fail due to unexpected application or hardware errors (but if it must fail, it does so gracefully). Such an operating system will provide services to each application unless the hardware those services requires fails to function.
- role** (RBAC)—Represents a set of tasks assigned to a member of an organization. Each role is assigned a set of privileges, which define the objects that users in each role can access.
- role-based access control (RBAC)**—Access control model in which users are assigned roles.
- roll**—See swap.
- roll back a transaction**—To return the system to the state that existed before the transaction was processed.
- root**—Beginning of a file system's organizational structure.
- root directory**—Directory that points to the various user directories.
- root key**—Used by the Internet Policy Registration Authority (IPRA) to sign certificates exclusively for policy creation authorities.
- root user** (Linux)—See superuser.
- rotational latency**—Time required for a disk to rotate a requested data item from its current position to a position adjacent to the read/write head.
- rotational optimization**—Disk scheduling technique that reduces access times by next servicing the request to the nearest sector in the read/write head's current cylinder.
- round-robin (RR) scheduling**—Scheduling policy that permits each *ready* process to execute for at most one quantum per round. After the last process in the queue has executed once, the scheduler begins a new round by scheduling the first process in the queue.
- round-robin job (RRJob) scheduling**—Job-aware process scheduling algorithm employing a global run queue in which jobs are dispatched to processors in a round-robin fashion.
- round-robin process (RRprocess) scheduling**—Job-blind multiprocessor scheduling algorithm that places each process in a global run queue and schedules these process in a round-robin manner.
- router**—Computer that is an intermediate destination between the sending host and the receiving host. The router is responsible for determining where to send a datagram next in order for it to eventually reach its destination.
- routing**—Determining the best route between two points and sending packets along this route.
- Routing Information Protocol (RIP)**—Protocol that defines how routing information is propagated throughout networks. RIP requires routers to share their entire routing table with other routers; this limits its use to small networks.
- routing table**—Representation of a network used to determine where routers should send datagrams next on their path to their destination.
- RSA**—Popular public-key algorithm, which was developed in 1977 by MIT professors Ron Rivest, Adi Shamir and Leonard Adleman.

run queue (Linux)—List of processes waiting to execute on a particular processor.

running state—Process (or thread) state in which a process (or thread) is executing on a processor.

S

safe asynchronous write (in NFS-3)—Allows a server to return before a write has been completed.

safe state—State of a system in Dijkstra's banker's algorithm in which there exists a sequence of actions that will allow every process in the system to finish without the system becoming deadlocked.

saturation — Condition of a resource that has no excess capacity to fulfill new requests.

saturation threshold—A level of resource utilization above which the resource will refuse access. Designed to reduce deadlock, it also reduces throughput.

scalability (scheduler)—Characteristic of a scheduler that ensures system performance degrades gracefully under heavy loads.

scalable operating system—Operating system that is able to employ resources as they are added to the system. It can readily adapt its degree of multiprogramming to meet the needs of its users.

SCAN disk scheduling—Disk scheduling strategy that reduces unfairness and variance of response times as compared to SSTF by servicing the request that requires the shortest seek distance in a preferred direction. SCAN behaves much like SSTF in terms of high throughput and good mean response times. However, because SCAN ensures that all requests in a given direction will be serviced before the requests in the opposite direction, it offers a lower variance of response times than SSTF. Also called the elevator algorithm.

scanner—See lexical analyzer.

scheduler activation—Mechanism that allows a user-level library to schedule kernel threads.

script-driven simulator—Simulator controlled by data carefully designed to reflect the anticipated environment of the simulated system; evaluators derive this data from empirical observations.

scriptlet—Java code embedded in a JSP.

second-chance page-replacement strategy—Variation of FIFO page replacement that uses the referenced bit and a FIFO queue to determine which page to replace. If the oldest page's referenced bit is off, second chance replaces the page; otherwise it turns off the referenced bit on the oldest page and moves it to the tail of the FIFO queue. If its referenced bit is on, the strategy turns off the bit and examines the next page or pages until it locates a page with its referenced bit turned off.

second extended file system (ext2fs) (Linux)—See ext2fs.

secret-key cryptography—Technique that performs encryption and decryption using the same secret key to encrypt and decrypt a message. The sender encrypts a message using the secret key, then sends the encrypted message to the intended recipient, who

decrypts the message using the same secret key. Also called symmetric cryptography.

secondary storage—Memory that typically stores large quantities of data persistently. Secondary storage is one level lower than main memory in the memory hierarchy. After a computer is powered on, information is shuttled between secondary storage and main memory so that program instructions and data can be accessed by a processor. Hard disks are the most common form of secondary storage.

sector—Smallest portion of a track that can be accessed by an I/O request.

sector queuing—See rotational optimization.

Secure Hash Algorithm (SHA-1)—Popular hash function used to create digital signatures; developed by NIST.

secure operating system—Operating system that prevents users and software from gaining unauthorized access to services and data.

Secure Sockets Layer (SSL)—Nonproprietary protocol developed by Netscape Communications that secures communication between two computers on the Internet.

Security Accounts Manager (SAM) (Windows XP)—Database that administers information about all security principals in the system. It provides services such as account creation, account modification and authentication.

security descriptor (Windows XP)—Data structure that stores information about which security principals may access a resource and what actions they may perform on that resource. The most important element of a security descriptor is its discretionary access control list (DACL).

security identifier (SID) (Windows XP)—Unique identification number assigned to each security principal in the system.

security mechanism—Method by which the system implements its security policy. In many systems, the policy changes over time, but the mechanism remains unchanged.

security model—Entity that defines a system's subjects, objects and privileges.

security patch—Code that addresses a security flaw.

security policy—Rules that govern access to system resources.

security principal (Windows XP)—Any user, process, service or computer that can perform an action in a Windows XP system.

Security Support Provider Interface (SSPI) (Windows XP)—Microsoft's standardized protocol for authentication and authorization recognized by both the Kerberos and NTLM authentication services.

seek operation — Operation that moves the disk head to a different cylinder.

seek optimization—Disk scheduling technique that reduces seek times by generally servicing requests to the cylinder near the read/write head.

seek time—Time it takes for the read/write head to move from its current cylinder to the cylinder containing the requested data record.

segment (TCP)—Piece of data sent by TCP. It includes the message and the TCP header.

- segment** (virtual memory)—Variable-size set of contiguous addresses in a process's virtual address space that is managed as one unit. A segment is typically the size of an entire set of similar items, such as a set of instructions in a procedure or the contents of an array, which enables the system to protect such items with fine granularity using appropriate access rights. For example, a data segment typically is assigned read-only or read/write access, but not execute access. Similarly, a segment containing executable instructions typically is assigned read/execute access, but not write access. Segments tend to create external fragmentation in main memory but do not suffer from internal fragmentation.
- segment descriptor** (IA-32 Intel architecture)—Segment map table entry that stores a segment's base address, present bit, limit address and protection bits.
- segment map table origin register**—Register that holds the location of a process's segment map table in main memory; having this information accessible in a high-speed register facilitates rapid virtual-to-physical address translation.
- segment-overflow exception**—Exception that occurs when a process attempts to access an address that is outside a segment.
- segment-protection exception**—Exception that occurs when a process attempts to access a segment in ways other than those specified by its access control mode (e.g., attempting to write to a read-only segment).
- segment selector** (IA-32 Intel architecture)—16-bit value indicating the offset into the segment map table at which the corresponding segment descriptor (i.e., segment map table entry) is located.
- selection evaluation**—Analysis regarding whether obtaining a computer system or application from a particular vendor is appropriate.
- selfish round-robin (SRR) scheduling**—Variant of round-robin scheduling in which processes age at different rates. Processes that enter the system are placed in a holding queue, where they wait until their priority is high enough for them to be placed in the active queue, in which processes compete for processor time.
- semaphore**—Mutual exclusion abstraction that uses two atomic operations (*P* and *V*) to access a protected integer variable that determines if threads may enter their critical sections.
- semaphore array** (Linux)—Linked list of semaphores that protect access to related resources.
- semaphore object** (Windows XP)—Synchronization object that allows a resource to be owned by up to a specified number of threads; it is essentially a counting semaphore.
- sender-initiated policy**—Dynamic load balancing policy in which overloaded processors attempt to find underloaded processors to which to migrate a process.
- separate kernels multiprocessor organization**—Scheme for delegating operating system responsibilities in which each processor executes its own operating system, but the processors share some global system information.
- seqlock** (Linux)—Mutual exclusion structure that combines a spin lock with a sequence counter. Seqlocks are used by interrupt handlers, which require immediate exclusive access to data.
- Sequenced Packet exchange (SPX)**—Novell Network's transport protocol, which offers connection-oriented services for IPX packets.
- sequential consistency**—Category of memory-coherence strategies in which coherence protocols are enforced immediately after a write to a shared memory location.
- sequential file organization**—File organization technique in which records are placed in sequential physical order. The "next" record is the one that physically follows the previous record.
- sequential write sharing**—Occurs when one client modifies a file, then another client tries to read or write its own cached copy of the file. Sequential write sharing introduces cache inconsistency.
- serial port**—Interface to a device that transfers one bit at a time (e.g. keyboards and mice).
- serialize**—To control access to a shared variable such that only one thread can access the variable at a time; another thread can access the variable only after the first has finished.
- serially reusable code**—Code that can be modified but is reinitialized each time it is used. Such code can be used by only one process or thread at a time.
- serially reusable resource**—See dedicated resource.
- serially reusable shared resource**—Resource that can be used by only one thread at a time.
- server**—Process that provides services to other processes (called clients). The machine on which these processes run is also called a server.
- Server Message Block (SMB)**—Network file sharing protocol used in Windows operating systems on top of which CIFS (Common Internet File System) is built.
- server stub**—A stub at the server side in RPC that prepares outbound data for transmission and translates incoming data so that it may be correctly interpreted.
- service rate**—Rate at which requests are completed by a resource.
- service ticket** (Kerberos)—Ticket that authorizes the client's access to specific network services.
- servlet**—Enhances the functionality of Web servers to provide capabilities such as secure access to Web sites, interacting with databases on behalf of a client, dynamically generating custom documents to be displayed by browsers and maintaining unique session information for each client.
- servlet container**—Server that executes a servlet. Also known as servlet engine.
- session key**—Secret key that is used for the duration of a transaction (e.g., a customer's buying merchandise from an online store).
- session layer** (in OSI)—Establishes, manages and terminates the communication between two end users.
- setup time**—Time required by a system operator and the operating system to prepare the next job to be executed.
- shadow page**—Block of data whose modified contents are written to a new block. Shadow pages are one way to implement transactions.

- shadow paging**—Transaction implementation that writes modified blocks to a new block. The copy of the block that is unmodified is released as free space when the transaction has been committed.
- shadow password file (UNIX)**—Protects the password file from crackers by storing information other than the encrypted passwords in the normal password file and storing the encrypted passwords in the shadow password file that can be accessed only by users with root privileges.
- shadowing (RAID)**—See mirroring.
- shared bus**—Multiprocessor interconnection scheme that uses a single communication path to connect all processors and memory modules.
- shared library**—Collection of functions shared between several programs.
- shared resource**—Resource that can be accessed by more than one process.
- shared virtual memory (SVM)**—An extension of virtual memory concepts to multiprocessor systems; SVM presents the illusion of shared physical memory between processors and ensures coherence for pages accessed by separate processors.
- shell**—Application (typically GUI or text based) that enables a user to interact with an operating system
- shortest-access-time-first (SATF) disk scheduling**—Disk scheduling strategy that next services the request that requires the shortest access time (i.e., positioning time plus transmission time). SATF exhibits higher throughput than SPTF, but large requests can be indefinitely postponed by a series of smaller requests, and requests to the innermost or outermost cylinders can be indefinitely postponed by requests to intermediate cylinders.
- shortest-latency-time-first (SLTF) disk scheduling**—Disk scheduling strategy that examines all of the waiting requests and services the one with the shortest rotational delay first. This strategy has been shown to be close to the theoretical optimum and is relatively easy to implement.
- shortest-positioning-time-first (SPTF) disk scheduling**—Disk scheduling strategy that next services the request that requires the shortest positioning time. SPTF results in high throughput and a low mean response time, similar to SSTF, and can also indefinitely postpone requests to the innermost and outermost cylinders.
- shortest-process-first (SPF) scheduling (multiprocessor)**—Job-blind multiprocessor scheduling algorithm, employing a global run queue, that selects the process with the smallest processor time requirement to execute on an available processor.
- shortest-process-first (SPF) scheduling (uniprocessor)**—Nonpreemptive scheduling algorithm in which the scheduler selects a process with the smallest estimated runtime-to-completion and runs the process to completion.
- shortest-remaining-time (SRT) scheduling**—Preemptive version of SPF in which the scheduler selects a process with the smallest estimated remaining runtime-to-completion.
- shortest-seek-time-first (SSTF) disk scheduling**—Disk scheduling strategy that next services the request that is closest to the read/write head's current cylinder (and thus incurs the shortest seek time), even if that is not the first one in the queue. By reducing average seek times, SSTF achieves higher throughput rates than FCFS, and mean response times tend to be lower for moderate loads. One significant drawback is that higher variances occur on response times because of the discrimination against the outermost and innermost tracks; in the extreme, starvation of requests far from the read/write heads could occur.
- signal**—Message sent by software to indicate that an event or error has occurred. Signals cannot pass data to their recipients.
- signal (semaphores)**—Operation on a semaphore that increments the value of the semaphore's variable. If threads are sleeping on the semaphore, the signal wakes one and decrements the semaphore's value by 1.
- signal-and-continue monitor**—Monitor that allows a thread to signal that the monitor is available, but does not require the thread to release the lock until it exits the monitor, at which point a signaled thread may enter the monitor.
- signal-and-exit monitor**—Monitor that requires a thread to release the lock on the monitor as soon as the thread signals another thread.
- signal handler**—Code that is executed in response to a particular signal type.
- signal mask**—Data structure that specifies which signals are not delivered to a thread. Depending on the signal type and default action, masked signals are either queued or dropped.
- signaled state (Windows XP)**—State in which a synchronization object can be placed, allowing one or more threads *waiting* on this object to awaken.
- signature-scanning virus detection**—Antivirus technique that relies on knowledge of virus code.
- Simple Object Access Protocol (SOAP)**—Messaging protocol for transporting information and instructions between Web services, using XML as a foundation for the protocol.
- simulation**—Performance evaluation technique in which an evaluator develops a computerized model of a system being evaluated. The model is then run to reflect the behavior of the system being evaluated.
- single-instruction-stream, multiple-data-stream (SIMD) computer**—Computer architecture consisting of one or more processing elements that execute instructions from a single instruction stream that act on multiple data items.
- single-instruction-stream, single-data-stream (SISD) computer**—Computer architecture in which one processor fetches instructions from a single instruction stream and manipulates a single data stream; this architecture describes traditional uniprocessors.
- single-level directory structure**—See flat directory structure.
- single sign-on**—Simplifies the authentication process by allowing the user to log in once, using a single password.
- single-stream batch-processing system**—Batch-processing system that places ready jobs in available partitions from one queue of pending jobs.

- Single UNIX Specification**—Specification (created by The Open Group) to which an operating system must conform to earn the right to display the UNIX trademark (see www.unix.org/version3/overview.html).
- single-use timer** (Windows XP)—Waitable timer object that is used once and then discarded.
- single-user contiguous memory allocation system**—System in which programs are placed in adjacent memory addresses and the system services only one program at a time.
- size** (file)—Amount of information stored in a file.
- skeleton**—Server-side stub.
- slab** (Linux)—Page of memory that reduces internal fragmentation due to small structures by storing multiple structures smaller than one page.
- slab allocator** (Linux)—Kernel entity that allocates memory for objects placed in the slab cache.
- slab cache** (Linux)—Cache that stores recently used slabs.
- slave node** (in a Beowulf cluster)—Beowulf cluster node that is not a head node.
- sleep interval**—Period of time (specified by the thread that is about to enter the *sleeping* state) during which a thread remains in the *sleeping* state.
- sleep/wakeup lock**—Mutual exclusion lock in which waiting processes block, and a releasing process wakes the highest-priority waiting process and gives it the lock.
- sleeping state**—Thread state in which a thread cannot execute until being returned to the *ready* state after the sleep interval expires.
- Small Computer Systems Interface (SCSI)**—Interface designed to support multiple devices and high-speed connections. The SCSI interface supports a larger number of devices than the less inexpensive IDE interface and is popular in Apple systems and computers containing large numbers of peripheral devices.
- smallest-number-of-processes-first (SNPF) scheduling**—Job-aware process scheduling algorithm, employing a global job-priority queue, where job priority is inversely proportional to the number of processes in a job.
- smart card**—Credit card size data store that serves many functions, including authentication and data storage.
- socket**—Interprocess communication mechanism that allows processes to exchange data by establishing direct communication channels. Enables processes to communicate over a network using read and write calls.
- socket** (network)—Software construct that represents one endpoint of a connection.
- socket address**—Unique identifier for a socket.
- socketpair** (Linux)—Pair of connected, unnamed sockets that can be used for bidirectional communication between processes on a single system.
- soft affinity**—Type of processor affinity in which the scheduling algorithm tries, but does not guarantee, to schedule a process only on a single node throughout its life cycle.
- soft link**—File that specifies the pathname corresponding to the file to which it is linked.
- soft real-time scheduling**—Scheduling policy that guarantees that real-time processes are scheduled with higher priority than non-real-time processes.
- softirq** (Linux)—Software interrupt handler that is reentrant and not serialized, so it can be executed on multiple processors simultaneously.
- software interrupt handler** (Linux)—Interrupt-handling code that can be performed without masking interrupts and can therefore be preempted.
- Solaris**—UNIX version based on both System V Release 4 and SunOS, designed by AT&T and Sun collaboratively.
- Solo**—Small operating system created by Per Brinch Hansen to demonstrate fail-safe concurrent programming.
- source code**—Program code typically written in a high-level language or assembly language that must be compiled or interpreted before it can be understood by a computer.
- source tree** (Linux)—Structure that contains source-code files and directories. Provides a logical organization to the monolithic Linux kernel.
- space-partitioning scheduling**—Multiprocessor scheduling strategy that attempts to maximize processor affinity by scheduling collaborative processes on a single processor (or single set of processors); the underlying assumption is that these processes will access the same shared data.
- space-time product**—Value that measures the product of a process's execution time (i.e., the duration for which a process occupies memory) and the amount of real-memory space the process occupies. Ideally, memory management strategies should reduce this quantity to increase a system's degree of multiprogramming.
- sparse file** (NTFS)—File with large blocks of regions filled with zeroes that NTFS tracks using a list of empty regions rather than explicitly recording each zero bit.
- spatial locality**—Empirical property that, in paging systems, states that processes tend to favor certain subsets of their pages, and that these pages tend to be near one another in a process's virtual address space. A process accessing sequential indices of an array exhibits spatial locality.
- spawning a process**—A parent process creating a child process.
- specified user** (file access control)—Identity of an individual user (other than the owner) that may use a file.
- SPECmark**—Standard benchmark for testing systems; SPECmarks are published by the **Standard Performance Evaluation Corporation (SPEC)**.
- speculative loading**—Technique whereby a processor retrieves from memory data specified by an instruction that has yet to be executed; when the instruction is executed, the processor performs a verifying load to ensure the data's consistency.
- spin lock**—Lock that provides mutually exclusive access to critical sections. When a process holding the lock is executing inside its critical section, any process concurrently executing on a different

- processor that attempts to acquire the lock before entering its critical section is made to busy wait.
- spindle**—Moving-head disk component that spins platters at high speeds.
- spool (simultaneous peripheral operations online)**—Method of I/O in which processes write data to secondary storage where it is buffered before being transferred to a low-speed device.
- Sprite**—Distributed operating system whose goal is transparency and complete consistency.
- stability**—Condition of a system that functions without error or significant performance degradation.
- stack region**—Section of process's address space that contains instructions and values for open procedure calls. The contents of the stack grow as a process issues nested procedure calls and shrink as called procedures return.
- Standard Application (SAP) benchmarks**—Popular benchmarks used to evaluate a system's scalability.
- Standard Performance Evaluation Corporation (SPEC)**—Organization that develops standard, relevant benchmarks (called SPECmarks), which are used to evaluate a variety of systems; SPEC publishes the results of tests with these benchmarks on real systems.
- standardized interface**—Allows each client/server pair to communicate using a single, common interface that is understood by both sides.
- Standby Page List (Windows XP)**—List of page frames that are consistent with their on-disk version and can be freed.
- standby state (Windows XP)**—Thread state denoting a thread that has been selected for execution.
- star network**—Network containing a hub that is directly connected to all other nodes in the network. The hub is responsible for relaying messages between nodes.
- starvation**—Situation in which a thread waits for an event that might never occur, also called indefinite postponement.
- starvation limit (Linux)**—Time at which high-priority processes are placed in the expired list to prevent low-priority processes from being indefinitely postponed.
- state information**—Data that describes the status of one or more resources.
- state transition**—Change of a process from one state to another.
- stateful server**—Keeps state information of the client requests—such as the file name, a pointer to the file and the current position in the file—so that the subsequent access to the file is easier and faster.
- stateless server**—The server does not keep state information of the client requests, so the client must specify which file to access in each request.
- static analysis**—Intrusion detection method which attempts to detect when applications have been corrupted by a hacker.
- static load balancing**—Category of load balancing algorithms that assign a fixed number of processors to a job when it is first scheduled.
- static priority level (Linux)**—Integer value assigned to a process when it is created that determines its scheduling priority.
- static RAM (SRAM)**—RAM that does not need to be refreshed and will hold data as long as it receives power.
- static real-time scheduling algorithm**—Scheduling algorithm that uses timing constraints to assign fixed priorities to processes before execution.
- steganography**—Technique that hides information within other information, derived from Latin roots meaning "covered writing."
- stream**—Sequence of objects fed to the processor.
- STREAM**—Popular synthetic benchmark which tests the memory subsystem.
- stream socket**—Socket that transfers data using the TCP protocol.
- strip (RAID)**—Smallest unit of data operated on by a RAID system. The set of strips at the same location on each disk is called a stripe.
- stripe (RAID)**—Set of strips at the same location on each disk in a RAID system. Striping enables RAID systems to access files using multiple disks at once, which improves transfer times.
- structured programming**—Disciplined approach to creating programs that are clear, correct and easy to modify.
- Structured Query Language (SQL)**—Database language that allows users to find data items that have certain properties, and also to create tables, specify integrity constraints, manage consistency and enforce security.
- stub**—prepares outbound data for transmission and translates incoming data so that it may be correctly interpreted
- stub/skeleton layer in RMI**—Contains parameter-marshaling structures analogous to the client and server stubs of RPC.
- substitution cipher**—Encryption technique whereby every occurrence of a given letter is replaced by a different letter. For example, if every "a" were replaced by a "b," every "b" by a "c," and so on the word "security" would encrypt to "tfdvjsjuz."
- sufficient conditions for deadlock**—The four conditions—mutual exclusion, no-preemption, wait-for and circular-wait—which are necessary and sufficient for deadlock.
- Sun Open Net Environment (Sun ONE)**—Consists of three components—a vision, an architecture and a conceptual model for developing standards-based software.
- superblock**—Block containing information critical to the integrity of the file system (e.g., the location of the file system's free block list or bitmap, the file system identifier and the location of the file system root).
- superscalar architecture**—Technique in which a processor contains multiple execution units so that it can execute more than one instruction in parallel per clock cycle.
- superuser (root user) (Linux)**—User that may perform restricted operations (i.e., those that may damage the kernel and/or the system).
- supervisor call**—Request by a user process to the operating system to perform an operation on its behalf (also called a system call).
- supervisor state**—See kernel mode.

- suspend/resume**—Method of halting a process, saving its state, releasing its resources to other processes, then restoring its resources after the other processes have released them.
- suspended state**—Process state (either *suspendedblocked* or *suspended-ready*) in which a process is indefinitely removed from contention for time on a processor without being destroyed. Historically, this operation allowed a system operator to manually adjust the system load and/or respond to threats of system failure.
- suspendedblocked state**—Process state resulting from the process being suspended while in the *blocked* state. Resuming such a process places it into the *blocked* state.
- suspendedready state**—Process state resulting from the process being suspended while in the *ready* state. Resuming such a process places it into the *ready* state.
- swap**—Method of copying a process's memory contents to secondary storage, removing the process from memory and allocating the freed memory to a new process.
- swap cache (Linux)**—Cache of page table entries that describes whether a particular page exists in the system swap file on secondary storage. If a page table entry is present in the swap cache, then its corresponding page exists in the swap file and does not need to be written to the swap file.
- swap instruction**—Operation that exchanges the values of two variables atomically. This instruction simplifies mutual exclusion implementations by eliminating the possibility that a thread will be preempted while performing a read-modify-write memory operation.
- switch**—Node that routes messages between component nodes.
- Symbian OS**—Small operating system for smart phones (mobile phones with the functionality of a PDA).
- symbol resolution**—Procedure performed by a linker that matches external references in one module to external names in another.
- symbol table**—Part of an object module that lists an entry for each external name and each external reference found in the module.
- symbolic name**—Device-independent name (e.g., a pathname).
- symmetric cryptography**—See secret-key cryptography.
- symmetric multiprocessor (SMP)**—Multiprocessor system in which processors share all resources equally, including memory, I/O devices and processes.
- symmetric policy**—Dynamic load balancing policy that combines the sender-initiated policy and the receiver-initiated policy to provide maximum versatility to adapt to environmental conditions.
- symmetrical multiprocessor organization**—Scheme for delegating operating system responsibilities in which each processor can execute the single operating system.
- synchronization**—Coordination between asynchronous concurrent threads to sequentialize their access to shared resources.
- synchronization segment (SYN)**—In TCP, the first handshaking segment sent; contains the sequence number of the source host.
- synchronization/acknowledgement segment (SYN/ACK)**—In TCP, the second handshaking segment sent; acknowledges that the SYN segment was received and contains the sequence number of the destination host.
- synchronized**—Java keyword that imposes mutual exclusive access to code inside an object.
- synchronous signal**—Signal generated due to execution of the currently *running* thread's instructions.
- syntax analyzer**—See parser.
- synthetic benchmark**—Another name for a synthetic program.
- synthetic program**—Artificial program used to evaluate a specific component of a system or constructed to mirror the characteristics of a large set of programs.
- SYSmark benchmark**—Popular benchmark for desktop systems developed by Business Application Performance Corporation (BAPCo).
- system call**—Procedure call that requests a service from an operating system. When a process issues a system call, the processor execution mode changes from user mode to kernel mode to execute operating system instructions that respond to the call.
- system file system (sysfs) (Linux)**—File system that allows processes to access structures defined by the unified device model.
- system penetration**—Successful breach of computer security by an unauthorized external user.
- system reaction time**—Time from when a job is submitted to a system until the first time slice of service is given to that job.
- system service (Windows XP)**—Process that executes in the background whether or not a user is logged into the computer and typically executes the server side of a client/server application; a system service for Windows XP is similar to a daemon for Linux.
- system tuning**—Process of making fine adjustments to a system based on performance monitoring to optimize the system's execution for a specific operating environment.
- system vector**—Vector containing the results of microbenchmarks for a number of operating system primitives for a specific system, used in an application-specific evaluation.
- system worker thread (Windows XP)**—Thread controlled by the system that sleeps until a kernel-mode component queues a work item for processing.
- systems programming**—Development of software to manage a system's devices and applications.

T

- table fragmentation**—Wasted memory consumed by block mapping tables; small blocks tend to increase the number of blocks in the system, which increases table fragmentation.
- take operation (in JavaSpaces)**—Removes from the JavaSpaces service an object that matches the given criteria. Take operations, together with write and read operations, allow distributed applications to dynamically exchange objects within JavaSpaces services.
- target computer (in process migration)**—Computer to which the process is migrated.
- task (Linux)**—User execution context (i.e., process or thread) in Linux.

- tasklet** (Linux)—Software interrupt handler that cannot be executed simultaneously on multiple processors. Tasklets are used to execute nonreentrant bottom halves of interrupt handlers.
- TCP window**—Flow-control and congestion-control mechanism in which only a certain amount of data can be sent by the network layer without the receiver explicitly authorizing the sender to send more.
- TCP/IP Protocol Stack**—Hierarchical decomposition of computer communications functions into four levels of abstraction called layers. These layers are the application layer, transport layer, network layer and link layer.
- teardown time**—Time required by a system operator and the operating system to remove a job from a system after the job has completed.
- temporal locality**—Property of events that are closely related over time. In memory references, temporal locality occurs when processes reference the same memory locations repeatedly within a short period.
- terminated state** (Windows XP)—Thread state that denotes a thread has finished executing.
- termination housekeeping**—In the case of mutual exclusion algorithms, task performed by the operating system to ensure that mutual exclusion is not violated and that threads can continue to execute if a thread terminates while executing its critical section.
- ternary relations**—Relation of degree 3.
- test-and-set**—Instruction implemented in hardware that atomically tests the value of a variable and sets the value of the variable to true. This instruction simplifies mutual exclusion implementations by eliminating the possibility that a thread will be preempted while performing a read-modify-write memory operation.
- text region**—Section of a process's address space that contains instructions that are executed by a processor.
- THE Multiprogramming System**—First layered operating system architecture, created by Edsger Dijkstra.
- thin client**—Application that requires minimal client-side support.
- thrashing**—Excessive paging activity causing low processor utilization that occurs when a process's memory allocation is smaller than its working set. This results in poor performance, as the process spends most of its time waiting as pages are transferred between secondary storage and main memory.
- thread**—Entity that describes an independently executable stream of program instructions (also called a thread of execution or thread of control). Threads facilitate parallel execution of concurrent activities within a process.
- thread environment block (TEB)** (Windows XP)—User-space data structure that stores information about a thread, such as the critical sections owned by a thread and exception-handling information.
- thread local storage (TLS)** (Windows XP)—Area of a thread's process's address space where the thread can store private data, inaccessible to other threads.
- thread pool** (Windows XP)—Collection of worker threads that sleep until a request is queued to them, at which time one of the threads awakens and executes the queued function.
- thread pooling**—Threading technique that employs a number of kernel threads that exist for the duration of the process that creates them. This technique can improve performance by reducing the number of costly thread creation and termination operations.
- thread state**—Status of a thread (e.g., *running*, *ready*, *blocked*, and so on).
- three-tier system**—System which offers a separation of the application logic, the user interface and the data. The user interface tier (also called the client tier) communicates with the user. The application logic tier is responsible for the logic associated with the system's function. The data tier stores the information that the user wishes to access.
- throughput**—Amount of work performed per unit time. Throughput can be measured as the number of processes that complete per unit time.
- thundering herd**—Phenomenon that occurs when many processes awaken when a resource becomes available; only one process acquires the resource, and the others test the lock's availability and reblock, wasting processor cycles.
- Ticket Granting Service (TGS)** (Kerberos)—Server that authenticates client's rights to access specific network services.
- Ticket-Granting Ticket (TGT)** (Kerberos)—Ticket returned by Kerberos's authentication server. It is encrypted with the client's secret key that is shared with the authentication server. The client sends the decrypted TGT to the Ticket Granting Service to request a service ticket.
- tightly coupled system**—System in which processors share most resources; these systems provide higher performance but are less fault tolerant and flexible than loosely coupled systems.
- time bomb**—Virus that is activated when the clock on the computer matches a certain time or date.
- time-of-day clock**—Clock that measures time as perceived outside of a computer system, typically accurate to thousandths or millionths of a second.
- time slice**—See quantum.
- time slicing**—Scheduling each process to execute for at most one quantum before preemption.
- time stamp**—Records the local time at which the message was sent.
- time stamping** (nonrepudiation)—Technique that binds a time and date to a digital document, which helps solve the problem of nonrepudiation.
- time-stamping agency**—Organization that digitally time stamps a document that has been digitally signed.
- timer-queue timer** (Windows XP)—Timer that signals a worker thread to perform a specified function at a specified time.
- timesharing scheduling**—Multiprocessor scheduling technique that attempts to maximize parallelism by scheduling collaborative processes concurrently on different processors.
- timesharing system**—Operating system that enables multiple simultaneous interactive users.
- timing**—Raw measure of an isolated hardware performance metric, such as a BIPS rating, used for quick comparisons between systems.

- timing constraint**—Time period during which a process (or subset of a process's instructions) must complete.
- TIPS (trillion instructions per second)**—Unit used to categorize the performance of a particular computer; a rating of one TIPS means a processor can execute one trillion instructions per second.
- tmpfs (temporary file system)** (Linux)—Similar to ramfs, but does not require formatting before use, meaning that the system can store files in the tmpfs without the organizational overhead typical of most file systems.
- token**—Empty frame used to ensure that only one host is transmitting data at a time in the Token Ring and FDDI protocols.
- token** (in compiling)—Characters in a program, separated by the lexical analyzer, that generally represent keywords, identifiers, operators or punctuation.
- token** (in token-based authentication)—Unique identifier for authentication.
- token-based authentication**—Authentication technique that issues a token unique to each session, enabling users to access specific applications.
- Token Ring**—Protocol in which a token is circulated around a ring network. Only one host can own the token at a time, and only its owner can transmit data.
- Tomcat**—Official reference implementation of the JSP and servlet standards.
- top half of an interrupt handler** (Linux)—Nonpreemptible portion of interrupt-handling code that performs the minimum work required to acknowledge an interrupt before transferring execution to the preemptible bottom-half handler.
- total ordering**—Ensures that all events are observed in the same order by all processes.
- trace**—Tiny electrically conducting line that forms part of a bus.
- trace** (performance evaluation)—Record of real system activity, which is executed on systems to test how the system handles a sample workload.
- track**—Circular region of data on a platter. Sequential file data is typically placed contiguously on one track to improve access time by reducing seeking activity.
- trailer**—Control information appended to the end of a data message.
- transaction**—Atomic, mutually exclusive operation that either completes or is rolled back. Modifications to database entries are often performed as transactions to enable high performance and reduce the cost of deadlock recovery.
- Transaction Processing Performance Council (TPC) benchmarks**—Popular benchmarks which target database systems.
- transaction transparency**—Allows a system to achieve consistency by masking the coordination among a set of resources.
- transaction wait-for graph (TWFG)**—Graph that represents processes as nodes and dependencies as directed edges, which is used for distributed deadlock detection algorithms.
- transceiver**—Hardware device that attaches an Ethernet node to the network transmission medium. Transceivers test a shared medium to see if it is available before transmitting data, and monitor the medium to detect a simultaneous transmission called a collision.
- transient virus**—Virus that attaches itself to a particular computer program. The virus is activated when the program is run and deactivated when the program is terminated.
- transistor**—Miniature switch that either allows or prevents current from passing to enable processors to perform operations on bits.
- transition state** (Windows XP)—Thread state denoting a thread that has completed a wait but is not yet ready to run because its kernel stack has been paged out of memory.
- transitional fault** (Windows XP)—Fault issued by the MMU when it tries to access a page that is in main memory, but whose page frame's status is set to standby, modified, or modified no-write.
- Transitional Page List** (Windows XP)—List of page frames whose data is in the process of being moved to or from disk.
- translation lookaside buffer (TLB)**—High-speed associative memory map that holds a small number of mappings between virtual page numbers and their corresponding page frame numbers. The TLB typically stores recently used page table entries, which improves performance for processes exhibiting locality.
- Transmission Control Protocol (TCP)**—Connection-oriented transmission protocol designed to provide reliable communication over unreliable networks.
- Transmission Control Protocol/Internet Protocol (TCP/IP)**—Family of protocols that provide a framework for networking on the Internet.
- transmission medium**—Material used to propagate a signal (e.g., optical fiber or copper wire).
- transmission time**—Time required for a data record to pass by the read/write head.
- transparency**—Hides the distribution aspects from users of a distributed system.
- Transport Driver Interface (TDI) driver** (Windows XP)—Network driver that exports a network interface to upper-level kernel-mode clients and interacts with low-level NDIS drivers.
- transport layer**—Set of protocols responsible for end-to-end communication of data in a network. This layer exists in both the TCP/IP model and the OSI model of network communication.
- transport layer** (in RMI)—Works with the RRL to send marshaled messages between the client and server in RMI.
- transposition cipher**—Encryption technique whereby the ordering of the letters is shifted. For example, if every other letter, starting with "s," in the word "security" creates the first word in the ciphertext and the remaining letters create the second word in the ciphertext, the word "security" encrypts to "scrt euuy."
- trap**—In the IA-32 specification, an exception generated by an error such as overflow (when the value stored by a register exceeds the capacity of the register). Also generated when program control reaches a breakpoint in code.
- tree network**—Hierarchical network that consists of multiple star networks. The hub of the first star network is the root of the tree.

Each node that this hub connects serves as a hub for another star network and is a root of a subtree.

Triple DES—Variant of DES that can be thought of as three DES systems in series, each with a different secret key that operates on a block. Also called 3DES.

triple indirect pointer—Pointer in an inode that locates a block of doubly indirect pointers.

Trojan horse—Malicious program that hides within a trusted program or simulates the identity of a legitimate program or feature, while actually causing damage to the computer or network when the program is executed.

TSS—Operating system designed by IBM in the 1960s that offered timesharing and virtual memory capabilities. Although it was never released commercially, many of its capabilities appeared in later IBM systems.

tunneling—Process of placing IPv6 datagrams in the body of IPv4 datagrams when communicating with routers that do not support IPv6.

tuple—Particular element of a relation.

turnaround time—Time from when a request is submitted until the system completes servicing it.

two-factor authentication—Authentication technique that employs two means to authenticate the user, such as biometrics or a smart card used in combination with a password.

two-tier system—A system in which the user interface resides on the client, the data resides on the server and the application logic lies on one or both of these components.

type (file)—Description of a file's purpose (e.g., an executable program, data or directory).

U

unblock—Remove a process from the *blocked* state after the event on which it was waiting has completed.

unblocked record—Record containing exactly one logical record for each physical record.

undivided coscheduling algorithm—Job-aware process scheduling algorithm in which processes of the same job are placed in adjacent locations in the global run queue, and processes are scheduled round-robin.

unicast address—IP address used to deliver data to a single host.

Unicode—Character set that supports international languages and is popular in Internet and multilingual applications.

unified device model (Linux)—Internal device representation that relates devices to device drivers, device classes and system buses. The unified device model simplifies power management and hot swappable device management.

uniform-memory-access (UMA) multiprocessor—Multiprocessor architecture that requires all processors to share all of main memory; in general, memory-access time is constant, regardless of which processor requests data, except when the data is stored in a processor's cache.

Uniform Naming Convention (UNC) format (Windows XP)—Format that specifies a file's pathname, including on which server and in which directory on that server the file is located.

Uniform Resource Identifier (URI)—Name that references a specific resource on the Internet.

Uniform Resource Locator (URL)—A URI used to access a resource in a common protocol such as HTTP and FTP. Consists of the protocol, host name, port and path of the resource.

UNIVAC 1 (UNIVersal Automatic Computer)—First computer to introduce a magnetic storage tape.

Universal Description, Discovery and Integration (UDDI)—Defines XML-based rules for building directories in which companies advertise themselves and their Web services.

universal serial bus (USB)—Serial bus interface that transfers data up to 480Mbps per second, can supply power to its devices and supports hot swappable devices.

universally unique identifier (UUID)—ID that is guaranteed, for all practical purposes, to be unique in the world. UUIDs uniquely identify an RPC interface.

UNIX—Operating system developed at Bell Laboratories that was written using the C high-level programming language.

unknown state (Windows XP)—Thread state denoting that the some error has occurred and the system does not know the state of the thread.

unmap a page (Linux)—To update page table entries to indicate that the corresponding page is no longer resident.

unreliable network—Network that may damage or lose packets.

unsafe state—State of a system in Dijkstra's banker's algorithm that might eventually lead to deadlock because there might not be enough resources to allow any process to finish.

unsigned state (Windows XP)—State in which a synchronization object can be; threads *waiting* on this object do not awaken until the object transitions to the *signaled* state.

update (file)—Operation that modifies an existing data item in a file.

update image (RAID)—Method to reduce parity computation time by storing the difference between new and old parities in memory instead of performing a read-modify-write cycle.

update set (in Coda)—Specifies, to each member of the AVSG, the members of the AVSG that have successfully performed the write.

usable operating system—Operating system that has the potential to serve a significant user base by providing an easy-to-use interface and supporting a large set of user-oriented applications.

user classes—Classification scheme that specifies individual users or groups of users that can access a file.

User Datagram Protocol (UDP)—Connectionless transmission protocol which allows datagrams to arrive out of order, duplicated or not at all.

user directory—Directory that contains an entry for each of a user's files; each entry points to where the corresponding file is stored on its storage device.

user-level threads—Threading model in which all threads in a process are assigned to one execution context.

user mode—Mode of operation that does not allow processes to directly access system resources.

user-mode APC (Windows XP)—APC queued by user-mode thread and executed by the target thread when the target thread enters an alertable *wait* state.

user-mode driver (Windows XP)—Device driver that executes in user space.

User-Mode Linux (UML) (Linux)—Linux kernel that executes as a user process within a host Linux system.

user state—See user mode.

utilization—Fraction of time that a resource is in use.

V

V operation—Operation on a semaphore that increments the value of the semaphore's variable if there are no threads waiting on the semaphore. If threads are waiting, the *V* operation wakes one of these.

Valid Page List (Windows XP)—List of page frames that are currently in a process's working set.

validate a model—To demonstrate that a computer model is an accurate representation of the real system the model is simulating.

variable-partition multiprogramming—Method of assigning partitions that are the exact size of the job entering the system.

variance in response times—Measure of how much individual response times deviate from the mean response time.

variant—Virus whose code has been modified from its original form, yet still retains its malicious payload.

vector-based methodology—Method of calculating an application-specific evaluation of a system based on the weighted average of the microbenchmark results for the target system's primitives; the weights are determined by the target application's relative demand for each primitive.

vector processor—Type of SIMD computer containing one processing unit that executes instructions that operate on multiple data items.

Venus (in AFS)—User-level process that interacts with the Vice processes run on distributed file servers to govern distributed file access.

very long instruction word (VLIW)—Technique in which a compiler chooses which instructions a processor should execute in parallel and packages them into a single (very long) instruction word; the compiler guarantees that there are no dependencies between instructions that the processor executes at the same time.

Vice (in AFS)—Entity that governs distributed file access in AFS.

virtual address—Address that a process accesses in a virtual memory system; virtual addresses are translated to real addresses dynamically at execution time.

Virtual Address Descriptor (VAD) (Windows XP)—Structure in memory that contains a range of virtual addresses that a process may access.

virtual address space—Set of memory addresses that a process can reference. A virtual address space may allow a process to reference more memory than is physically available in the system.

virtual file system (Linux)—Interface that provides users with a common view of files and directories stored across multiple heterogeneous file systems.

virtual file system (VFS) (in distributed systems)—Provides the abstraction of a common file system at each client and is responsible for all distributed file operations.

virtual machine—Application that emulates the functionality of a computer system. A virtual machine can execute applications that are not directly compatible with the physical system that runs the virtual machine. The user "sees" the computer not as the virtual machine, but as the underlying physical machine.

virtual machine operating system—Software that creates the virtual machine.

virtual memory—Capability of operating systems that enables programs to address more memory locations than are actually provided in main memory. Virtual memory systems help remove much of the burden of memory management from programmers, freeing them to concentrate on application development.

virtual memory area (Linux)—Structure that describes a contiguous region of a process's virtual address space so that the kernel can perform operations on this region as a unit.

virtual memory manager (VMM) (Windows XP)—Executive component that manages virtual memory.

Virtual Private Network (VPN)—Technique that securely connects remote users to a private network using public communication lines. VPNs are often implemented using IPsec.

virus—Executable code (often sent as an attachment to an e-mail message or hidden in files such as audio clips, video clips and games) that attaches to or overwrites other files to replicate itself, often harming the system on which it resides.

virus signature—Segment of code that does not vary between virus generations.

VM operating system—One of the first virtual machine operating systems; developed at IBM in the 1960s and still used widely today; its latest version is the **z/VM**.

VMS—Operating system for the DEC VAX computers, designed by David Cutler's team.

volatile storage—Storage medium that loses data in the absence of power.

volatility (file)—Frequency with which additions and deletions are made to a file.

volume—Unit of storage that may hold multiple files.

volume (in AFS-2)—Introduced in AFS-2 to manage subtrees. Volumes are primarily of administrative value, allowing replication and isolation of certain subtrees, and are therefore transparent to users.

volume storage group (VSG) (in Coda)—Volumes are logical pieces of the file system and are replicated physically across multiple file servers. Servers that hold the same volume are known as a volume storage group (VSG).

voluntary page release—Occurrence when a process explicitly releases a page frame that it no longer needs. This can improve performance by reducing the number of unused page frames allocated to a process, leaving more memory available.

W

wait (semaphores)—If the variable in the semaphore is 0, then the operation blocks the calling thread. If the variable is greater than 0, the operation will decrement the variable by one and allow the calling thread to proceed. Wait is also called the *P* operation.

wait-die deadlock prevention strategy—Prevents deadlock by denying the wait-for condition. Assigns individual processes unique priorities based on when they were created. A process will wait if it was created after the process it is waiting on. A process will die if it was created before the process it is waiting on.

wait-for condition—One of the four necessary conditions for deadlock; states that deadlock can occur only if a process is allowed to wait for a resource while it holds another.

wait function (Windows XP)—Function called by a thread to wait for one or more dispatcher objects to enter a *signaled* state; calling this function places a thread in the *waiting* state.

wait queue—See wait set.

wait set—In Java, a set of threads waiting to reacquire the lock on a monitor.

waitable timer object (Windows XP)—Synchronization object that becomes *signaled* after a specified amount of time elapses.

waiting state—Thread state from which a thread cannot execute until transitioning to the *ready* state via a wake or notify operation.

wake—Thread operation that transitions its target from the *waiting* state to the *ready* state.

wall clock time—Measure of time as perceived by a user.

Web-based Distributed Authoring and Versioning (WebDAV) (Windows XP)—Network file sharing protocol that allows users to write data directly to HTTP servers and is designed to support collaborative authoring between groups in remote locations.

Web defacing—Attack that maliciously modifies an organization's Web site.

Web service methods (in .NET)—Methods contained in a .NET Web service class.

Web services—Set of services and related standards that can allow any two computer applications to communicate and exchange data over the Internet. Web services operate using open, text-based standards that enable components written in different languages and on different platforms to communicate. They are ready-to-use pieces of software on the Internet.

Web Services Description Language (WSDL)—Provides a standard method of describing Web services and their specific capabilities.

WebMark—Popular benchmark for Internet performance developed by Business Application Performance Corporation (BAPCo).

WebNFS—Allows NFS clients to access WebNFS-enabled servers with a minimum of protocol overhead. Marketed as the file system for the Web, WebNFS is designed to improve NFS functionality and performance over wide area Internet and intranets.

Whetstone—Classic synthetic program which measures how well systems handle floating point calculations, and has thus been helpful in evaluating scientific programs.

wide area network (WAN)—Type of network connecting two or more local area networks, usually operating over great geographical distances. WANs are generally implemented with a mesh topology and high-capacity connections. The largest WAN is the Internet.

Wi-Fi Protected Access (WPA)—Wireless security protocol intended to replace WEP by providing improved data encryption and by enabling user authentication.

Win32 environment subsystem (Windows XP)—User-mode process interposed between the executive and the rest of user space that provides a typical 32-bit Windows environment.

Win32 service (Windows XP)—See system service (Windows XP).

Win32 thread—Threads natively supported in the Microsoft 32-bit Windows line of operating systems.

WinBench 99—Popular synthetic program used extensively today in testing a system's graphics, disk and video subsystems in a Microsoft Windows environment.

window manager (Linux)—Application that controls the placement, appearance, size and other attributes of windows in a GUI.

Windows API—Microsoft's interface for applications that execute in a Windows environment. The API enables programmers to request operating system services, which free the application programmer from writing the code to perform these operations and enables the operating system to protect its resources.

Windows Driver Model (WDM) (Windows XP)—Standard driver model that enables source-code compatibility across all Windows platforms; each WDM driver must be written as a bus driver, function driver or filter driver and support PnP, power management and WMI.

Windows Management Instrumentation (WMI) (Windows XP)—Standard that describes how drivers provide measurement and instrumentation data to users (e.g., configuration data, diagnostic data or custom data) and allow user applications to register for WMI driver-defined events.

Windows sockets version 2 (Winsock 2)—Adaptation and extension of BSD sockets for the Windows environment.

Windows XP Embedded—Embedded operating system, which uses the same binary files as Windows XP but allows designers to choose only those components applicable for a particular device.

WinHTTP (Windows XP)—API that supports communication between clients and servers over an HTTP session.

WinINet (Windows XP)—API that allows applications to interact with FTP, HTTP and Gopher protocols to access resources over the Internet.

- Wired Equivalent Privacy (WEP)**—Wireless security protocol that encrypts transmitted data and prevents unauthorized access to the wireless network.
- word**—Number of bits a system's processor(s) can process at once. In the data hierarchy, words are one level above bytes.
- worker thread**—Kernel thread that is a member of a thread pool. Worker threads may be mapped to any user thread in the process that created its thread pool.
- working directory**—Directory that contains files that a user can access directly.
- working set**—A program's favored subset of pages in main memory. Given a working set window, w , the process's working set of pages $W(t, w)$, is defined as the set of pages it references during the process-time interval $t - w$ to t .
- working set (Windows XP)**—All of the pages in main memory that belong to a specific process.
- working set maximum (Windows XP)**—Upper limit on the number of pages a process may have simultaneously in main memory.
- working set minimum (Windows XP)**—Number of pages in main memory the working set manager leaves a process when it executes the page-trimming algorithm.
- working set theory of program behavior**—Theory presented by Denning, which asserts that for a program to run efficiently, the system must maintain that program's favored subset (i.e., its working set) of pages in main memory. Given a working set window, w , the process's working set of pages $W(t, w)$, is defined as the set of pages referenced by the process during the process-time interval $t - w$ to t . Choosing the window size, w , is a crucial aspect of implementing working set memory management.
- working set window size**—Value that determines how far into the past the system should consider to determine what pages are in the process's working set.
- workload**—Measure of the amount of work that has been submitted to a system; evaluators determine typical workloads for a system and evaluate the system using these workloads.
- workstation login scripts**—Simple form of single sign-on in which users log in at their workstations, then choose applications from a menu.
- World Wide Web (WWW)**—Collection of hyperlinked documents accessible via the Internet using the Hypertext Transfer Protocol (HTTP). Web documents are typically written in languages such as HyperText Markup Language (HTML) and Extensible Markup Language (XML).
- worm**—Executable code that spreads and infects files over a network. Worms rarely require any user action to propagate, nor do they need to be attached to another program or file to spread.
- worst-fit strategy**—Memory placement strategy that places an incoming job in the largest hole in memory.
- wound**—When a process is wounded by another process, it will be rolled back.
- wound-wait deadlock prevention strategy**—Prevents deadlock by denying the no-preemption condition. Assigns individual processes unique priorities based on when they were created. A process requesting a resource held by another process will wound that process if the first one was created before the other. A process will wait if it was created after the process it is waiting on.
- write (file)**—Operation that outputs a data item from a process to a file.
- write access (virtual memory)**—Access right that enables a process to modify the contents of a page or segment; see also execute access, read access and append access.
- write access (file)**—Permission to access a file for writing.
- write-back caching**—Technique that writes buffered data to disk periodically, enabling the operating system to batch multiple I/Os that are serviced using a single request, which can improve system performance.
- write broadcast**—Technique for maintaining memory coherence in which the processor that performs a write broadcasts the write throughout the system.
- Write-Once, Read-Many (WORM) medium**—Storage medium that can be modified only once, but whose contents can be accessed repeatedly.
- write operation (in JavaSpaces)**—Operation that adds an object into the JavaSpaces serviceA
- write-through caching**—Technique that writes data both to the disk cache buffer and to disk each time cached data is modified. This technique prevents the system from batching requests, but reduces the possibility of inconsistent data in the event of a system crash.
- write-through mode (Windows XP)**—Method of writing to a pipe whereby write operations do not complete until the data being written is confirmed to be in the buffer of the receiving process.
- ## X
- XOR (exclusive-or) operation**—Operation on two bits that returns 1 if the two bits are not the same, 0 otherwise. RAID levels 3-5 use the XOR operation to generate parity bits.
- ## Z
- z/OS**—IBM operating system for zSeries mainframes and the latest version of MVS.
- Zeroed Page List (Windows XP)**—List of page frames whose bits are all set to zero.
- zone (memory) (Linux)**—Region of physical memory. Linux divides main memory in the low, normal and high zones to allocate memory according to the architectural limitations of a system.
- zone allocator**—Memory subsystem that allocates pages from the zone to which it is assigned.

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